Non-Stationary Structural Causal Bandits

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Abstract

We study the problem of sequential decision-making in environments governed by evolving causal mechanisms. Prior work on structural causal bandits—formulations that integrate causal graphs into multi-armed bandit problems to guide intervention selection—has shown that leveraging the causal structure can reduce unnecessary interventions by identifying possibly-optimal minimal intervention sets (POMISs). For complete intervention plans fixed before the episode, existing static methods can account for long-term effects by using the aggregate reward $R = \sum_t Y_t$ while keeping all original rewards non-manipulable. In this work, we propose a non-stationary structural causal bandit framework that leverages temporal structural causal models to capture evolving dynamics over time. We characterize how interventions propagate over time by developing graphical tools and assumptions, which form the basis for identifying non-myopic intervention strategies. Within this framework, we devise POMIS$^+$, which combines a current-slice POMIS with admissible support for a specified future model family and intervention set. The original future-border restriction can exclude a cumulative-reward optimum. The corrected aggregate static reduction preserves the best fixed-plan value under finite action domains, but does not establish exact POMIS$^+$ enumeration or adaptive-policy optimality. Our framework provides a principled way to reason about temporally-aware interventions by explicitly modeling information propagation across time. Published empirical results for the original temporal construction are retained; the replacement algorithm has not been evaluated by rerunning those experiments.

1 Introduction

The Multi-Armed Bandit (MAB) problem is a classic decision-making scenario where an agent sequentially selects actions (arms), each with an unknown but fixed reward distribution, to maximize cumulative reward [Sutton and Barto, 2018]. A common assumption in MAB formulations is that arms are independent, meaning that the reward distribution of one arm does not depend on another [Lai and Robbins, 1985, Auer et al., 2002a]. However, in many real-world settings, actions are not independent—hidden factors may simultaneously influence both the choice of action and the observed reward, introducing unobserved confounders (UCs). Bareinboim et al. [2015], Forney et al. [2017], Lattimore et al. [2016], Zhang and Bareinboim [2017] demonstrate that in the presence of UCs, observed rewards may not accurately reflect the true effect of actions when passive data are used. Confounding alone does not invalidate a randomized bandit algorithm evaluated against the best fixed interventional arm. To address this, Lee and Bareinboim [2018] use structural causal models (SCMs) [Pearl, 2009] and causal diagrams to identify which variables an agent should intervene on to learn an optimal policy.

In many real-world applications, the reward distribution associated with an action is not fixed but changes over time due to shifting contexts, user behavior or environmental factors. Such settings

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motivate the study of non-stationary multi-armed bandit (NS-MAB) problems, which extend the classical formulation to handle temporal variation in reward dynamics. This dynamic formulation requires agents to continuously adapt to evolving environments by incorporating temporal information into their decision-making process. In particular, it relaxes the assumption of fixed reward distributions that underpins the classical MAB setting. While prior approaches to NS-MAB address temporal shifts in reward through statistical modeling—such as bounded change models [Auer et al., 2002b] or recharging payoffs [Papadigenopoulos et al., 2022]—they typically model non-stationarity in purely statistical terms, overlooking the underlying causal mechanisms responsible for reward shifts. Yet, in many sequential decision-making problems, such shifts are driven by latent and evolving causal mechanisms. In such cases, ignoring the structural causes behind the reward changes can limit an agent's ability to make informed interventions. In contrast, we frame non-stationarity through the lens of SCMs, allowing us to reason about how interventions propagate over time. The details are provided in §E.

Related causal approaches to non-stationary MAB are discussed in §K. One could consider adapting existing causal methods for stationary settings. In particular, Lee and Bareinboim [2019] propose a latent projection-based method to identify possibly-optimal arms under partial causal knowledge (as detailed in §J.1). Time labels can be retained under that projection, which preserves interventional rewards for paired original and projected SCMs. Its limitations depend on which variables and constraints are retained. This matters because, in non-stationary environments, identifying truly optimal strategies—those informed by both causal structure and temporal dynamics—is crucial. It enables more effective and timely interventions in real-world settings such as healthcare, education, and resource allocation, where decisions must often be made under uncertainty and limited budgets. Intervening without regard to temporal structure may appear optimal in the short term but lead to sub-optimal cumulative outcomes, ultimately limiting long-term performance. By explicitly modeling temporal structure, our approach overcomes this by capturing how causal effects unfold over time. This enables non-myopic strategies that align with long-term reward maximization, which is essential for planning in dynamic environments.

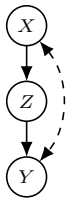
In this work, we approach the NS-MAB problem through the causal lens by introducing a structural framework that captures temporal dynamics via a rolled-out causal graph[2] (§3). We propose POMIS$^+$ that combines a current-slice POMIS with admissible future support under the stated local conditions (§5). We provide a graphical formulation of POMIS$^+$ that characterizes how interventions propagate across time, with a conditional local composition rule (§6). For complete fixed plans, we apply the existing static SCM-MAB method to an aggregate reward to preserve the cumulative optimum (§7). We retain the empirical evaluation of the original POMIS$^+$ construction across three non-stationary tasks; these curves have not been regenerated for the replacement algorithm (§8). §A outlines the paper's structure.

Contributions First, we propose a framework—non-stationary structural causal bandit (NS-SCM-MAB)—that combines the nature of NS-MAB with the underlying SCMs. Our framework models temporal dynamics not only in the reward distributions but also in the underlying causal structure, capturing how interventions propagate over time. This formulation enables temporally-aware decision-making, where an intervention at time t may have effects on both immediate and future outcomes. Second, to support this, we devise POMIS$^+$, which combines a current-slice POMIS with admissible support for a specified future model family and intervention set. Third, the partial-assignment result concerns the best finite boundary context for one fixed intervention set. Finally, the aggregate static reduction preserves the optimal value of complete fixed plans under the conditions in §7. This applies existing static results; it does not establish exact POMIS$^+$ enumeration, exact minimality within the deterministic-sum subclass or optimality of history-dependent policies. The experiments retain reported results for the original temporal method.

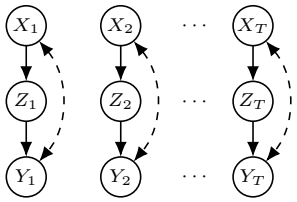
2 Preliminaries

We introduce notation from causal inference to understand multi-outcome causal MAB problems. Capital letters denote a single variable, and the domain of X is denoted $\mathscr{D}(X)$. Bold capital $\mathbf{X} =$

[2]A rolled-out causal graph is a time-unfolded representation of an SCM that explicitly captures temporal dependencies across time steps. Similar constructions appear in causal modeling over time, see e.g., Koller and Friedman [2009, p. 203].



(a) Stationary SCM-MAB



(b) Rolled-out stationary MAB

Figure 1: (a) shows the causal diagram for stationary SCM-MAB. The graph notation for a rolled-out version of the stationary MAB is shown in (b) indexing each round by its corresponding time step.

$\{X_1, \ldots, X_n\}$ represents a set of variables. Additionally, lowercase $x \in \mathscr{D}(X)$ represents a value of variable X, and the set of values is described as bold lowercase $\mathbf{x} \in \mathscr{D}(\mathbf{X}) = \times_{X \in \mathbf{X}} \mathscr{D}(X)$. We denote $\mathbf{x}[\mathbf{W}]$ as the values of $\mathbf{x}$ corresponding to the intersection $\mathbf{W} \cap \mathbf{X}$.

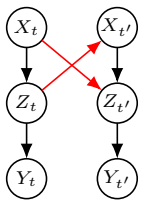
We adopt the structural causal model (SCM) [Pearl, 2009] framework. An SCM $\mathcal{M}$ is a tuple $\langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$, where $\mathbf{V}$ is a set of endogenous variables and $\mathbf{U}$ is a set of exogenous variables. Each $f_V \in \mathcal{F}$ is a function that determines each endogenous variable V. That is, $V \leftarrow f_V(\mathbf{pa}_V, \mathbf{u}_V)$ where $\mathbf{Pa}_V \subseteq \mathbf{V} \setminus \{V\}$ and $\mathbf{U}_V \subseteq \mathbf{U}$. $P(\mathbf{U})$ is a joint distribution over the exogenous variables. The SCM induces a causal diagram $\mathcal{G}$, which includes directed edges encoding functional relationships between endogenous variables, and bi-directed edges encoding UCs. We adopt family relationships $Pa(\cdot)_\mathcal{G}$, $Ch(\cdot)_\mathcal{G}$, $An(\cdot)_\mathcal{G}$, and $De(\cdot)_\mathcal{G}$ to denote parents, children, ancestors and descendants of a given variable where ancestors and descendants include its argument.

We consider a discrete-time setting where each time step $t \in \{1, \ldots, T\}$ (abbreviated $t \in [T]$) corresponds to a distinct point in time. We denote by $\mathbf{V}_t \subseteq \mathbf{V}$ a set of variables at each time step t, focusing on the specific variables indexed by the subscript t. Let $\mathbf{U}_t \subseteq \mathbf{U}$ be the set of exogenous variables, including exogenous variable $\mathbf{U}_{V_t}$ for every $V_t \in \mathbf{V}_t$ and unobserved confounders for every $V_t^i, V_t^j \in \mathbf{V}_t$ if $\mathbf{U}_{V_t^i}$ and $\mathbf{U}_{V_t^j}$ are correlated. $\mathbf{V}_{>t} \subseteq \mathbf{V}$ is the set of variables whose time index is strictly greater than t. Let $\mathbf{X}_t \subseteq \mathbf{V}_t \setminus \{Y_t\}$ denote the set of manipulative variables, where Y_t is the outcome variable. Unless stated otherwise, every non-reward variable is manipulable; the set of non-manipulable variables is $\mathbf{N} = \mathbf{Y} = \{Y_1, \ldots, Y_T\}$. Any further intervention restrictions must be specified separately. We use blackboard bold letter $\mathbb{V} = \{\mathbf{V}_1, \ldots, \mathbf{V}_T\}$ to represent the collection of all time-indexed variables throughout the sequence and use $\mathbb{F} = \{\mathcal{F}_1, \ldots, \mathcal{F}_T\}$ to denote the collection of all time-specific functions.

A vertex-induced subgraph is represented by $\mathcal{G}[\mathbf{V}']$ where $\mathbf{V}' \subseteq \mathbf{V}$. Given a causal diagram $\mathcal{G}$, a time-specific subgraph can be constructed as $\mathcal{G}[\bigcup_{i=t}^{t+\tau-1} \mathbf{V}_i]$, where t denotes the starting time step and τ represents the length of the time window. In particular, when $\tau = 1$, we refer to this graph as a 'time slice', denoted by $\mathcal{G}[\mathbf{V}_t]$. The probability of $Y = y$, when variables $\mathbf{X}$ are fixed to $\mathbf{x}$, is denoted $P(y \mid \text{do}(\mathbf{X} = \mathbf{x}))$ using the do-operator—an intervened probability. The graphical representation of the intervention is denoted $\mathcal{G}_{\overline{\mathbf{X}}}$; a mutilated graph where the incoming edges onto $\mathbf{X}$ are removed.

3 Non-Stationary Multi-Armed Bandit as a Structural Causal Model

We define the NS-SCM-MAB and identify the problems that arise when an existing *stationary* SCM-MAB solution is applied to the NS-MAB problem by reviewing the SCM-MAB [Lee and Bareinboim, 2018]. Let $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ be an SCM, and $Y \in \mathbf{V}$ be a reward variable, where $\mathscr{D}(Y) \subseteq \mathbb{R}$. The arms of the bandits are defined by the possible values $\mathbf{x}$ of the manipulative variable set $\mathbf{X}$, where $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$, and $\mathbf{x} \in \mathscr{D}(\mathbf{X})$. Each arm is associated with a reward distribution $P(Y \mid \text{do}(\mathbf{X} = \mathbf{x}))$. The expected reward of an arm is denoted by $\mathbb{E}[Y \mid \text{do}(\mathbf{X} = \mathbf{x})]$, written as $\mu_\mathbf{x}$. The optimal value of $\mathbf{x}$ for the expected reward is denoted by $\mathbf{x}^*$, i.e., $\mathbf{x}^* = \text{argmax}_{\mathbf{x} \in \mathscr{D}(\mathbf{X})} \mu_\mathbf{x}$. For clarity, we represent the information given to an



(a) $\mathcal{G}$

$$
\begin{aligned}
f_{X_t} &= U_{X_t} \\
f_{Z_t} &= U_{Z_t} \oplus x_t \\
f_{Y_t} &= U_{Y_t} \oplus z_t \\
f_{X_{t'}} &= U_{X_{t'}} \oplus z_t \\
f_{Z_{t'}} &= U_{Z_{t'}} \oplus x_{t'} \oplus x_t \\
f_{Y_{t'}} &= U_{Y_{t'}} \oplus z_{t'}
\end{aligned}
$$

(b) $\mathcal{F}$

Figure 2: SCM with non-stationary structure.

agent interacting with an SCM-MAB as $[\![\mathcal{G}, Y]\!]$. We now discuss time-expanded notation. When we expand it to the time horizon, for every $t \in [T]$, the causal structure among time-indexed variables is identical across all time steps (illustrated in Fig. 1(b)). The arms of the bandits are defined by the possible values $\mathbf{x}_t$ of the set $\mathbf{X}_t \subseteq \mathbf{V}_t \setminus \{Y_t\}$ and $\mathbf{x}_t \in \mathscr{D}(\mathbf{X}_t)$. The reward distribution is defined $P(Y_t \mid \mathrm{do}(\mathbf{X}_t = \mathbf{x}_t), \mathbb{1}_{t>1} \cdot I_{1:t-1})$ where $I_{1:t-1} = \mathrm{do}(\{\mathbf{X}_i = \mathbf{x}_i\}_{i=1}^{t-1})$ denotes previous interventions and $\mathbb{1}_{t>1}$ is the indicator function. We denote $\mathbb{E}[Y_t \mid \mathrm{do}(\mathbf{X}_t = \mathbf{x}_t), \mathbb{1}_{t>1} \cdot I_{1:t-1}]$ as $\mu_{\mathbf{x}_t, I_{1:t-1}}$, abbreviated as $\mu_{\mathbf{x}_t}$. The optimal value for $\mathbf{X}_t$ is denoted $\mathbf{x}_t^*$.

Non-stationarity in structural causal models Before defining *non-stationarity*, we clarify our understanding of *stationarity*. In the context of bandits, stationarity means that the reward distribution induced from pulling one arm (strategy) among all possible arms remains constant across all time steps $t \in [T]$. Interpreted from the perspective of the SCM-MAB, this implies that the distribution over the outcome variable Y—under interventions on any variables in the set of action nodes $\mathbf{X} \subseteq \mathbf{V} \setminus Y$—remains unchanged, regardless of earlier interventions, as discussed in §D. The comparison uses corresponding variables and the same intervention values across slices. Formally, we can express this as follows: for every pair of time steps $t, t' \in [T]$ with $t \neq t'$, a reward distribution $P(y_t \mid \mathrm{do}(\mathbf{x}_t), \mathbb{1}_{t>1} \cdot I_{1:t-1}) = P(y_{t'} \mid \mathrm{do}(\mathbf{x}_{t'}), \mathbb{1}_{t'>1} \cdot I_{1:t'-1})$. From this point of view, we introduce the NS-SCM-MAB where the reward distribution evolves over time.

Definition 3.1 (Non-stationary SCM-MAB)**.** Given a causal diagram $\mathcal{G}$, let $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ be an SCM and $\mathbf{Y} \subset \mathbf{V}$ be a set of reward variables $\mathbf{Y} = \bigcup_{i=1}^{T} \{Y_i\}$, where $T > 1$. An SCM-MAB $\langle \mathcal{M}, \mathbf{Y} \rangle$ is said to be non-stationary if, for some $t, t' \in [T]$ with $t < t'$, a corresponding arm $(\mathbf{X}, \mathbf{x})$ and admissible preceding intervention histories $I_{<t}, I'_{<t'}$ give different reward distributions:

$$P^{\mathcal{M}}(Y_t \mid do(\mathbf{X}_t = \mathbf{x}), I_{<t}) \neq P^{\mathcal{M}}(Y_{t'} \mid do(\mathbf{X}_{t'} = \mathbf{x}), I'_{<t'}).$$

Here $I_{<1}$ is empty, all displayed histories are interventions rather than observations, and reward domains are identified across slices. The disparity between reward distributions is referred to as a *reward distribution shift*.

The NS-SCM-MAB addresses the problem of arm selection in environments where the interventional distribution over reward variables may shift over time. This is illustrated in Fig. 2. As shown in the diagram, the value from the previous time step (t) can influence the subsequent time step (t'), as indicated by the red edges. This temporal influence causes the reward distribution to change over time, reflecting the non-stationary nature of the underlying SCM. To understand this setting from the agent's perspective, it is important to recognize that the topological ordering of variable generation is subordinated to the temporal ordering—variables associated with earlier time steps are always instantiated before those at later time steps. Based on this temporal structure, we define the notion of a temporal model within an SCM. A temporal model is an SCM that captures the time-specific causal mechanisms as perceived by the agent at a given time step.

Definition 3.2 (Temporal Model)**.** Let $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ be an SCM. For some $t \in [T]$ with $T > 1$, we define values $\mathbf{v}_t^\star$ and $\mathbf{u}_t^\star$, which correspond to the values of $\mathbf{V}_t^\star = Pa(\mathbf{V}_t) \setminus \mathbf{V}_t$ and the set of correlated variables $\mathbf{U}_t^\star$ between $V_t \in \mathbf{V}_t$ and $V_{t'} \in \mathbf{V}_{t'}$ for every $t \neq t'$, respectively. A temporal model $\mathcal{M}_t \mid \mathbf{v}_t^\star, \mathbf{u}_t^\star$ is defined as $\langle \mathbf{U}_t, \mathbf{V}_t, \mathcal{F}_t, P(\mathbf{U}_t \mid \mathbf{u}_t^\star) \rangle$ where each $f_{V_t} \in \mathcal{F}_t$ is a function that determines V_t given predetermined values $\mathbf{v}_t^\star$ and $\mathbf{u}_t^\star$. That is, $V_t \leftarrow f_{V_t}(\mathbf{pa}_{V_t}[\mathbf{V}_t] \cup \mathbf{v}_t^\star, \mathbf{u}_{V_t} \cup \mathbf{u}_t^\star)$ where $\mathbf{Pa}_{V_t} \subseteq \mathbf{V} \setminus \{V_t\}$ and $\mathbf{U}_{V_t} \subseteq \mathbf{U}_t$.

Concisely, we denote $\mathcal{M}_t \mid \mathbf{v}_t^\star, \mathbf{u}_t^\star$ by $\mathcal{M}_t$. Since the agent can *only* manipulate variables within the current time step, the temporal model is regarded as an underlying mechanism from the agent's perspective for each time step. Fixed latent boundary values in this construction are a mathematical device and are not assumed observable by the agent. A realized boundary is generally random under a fixed intervention history; evaluating an unconditional reward requires averaging over its induced law, as in Prop. K.1. The key point is that the mechanism may vary across time steps, which arises from *information propagation*, resulting in a reward distribution shift. A *myopic* agent, however, fails to adapt to this change and follows a strategy which chooses actions based on the current information only.

Non-stationarity caused by information propagation We introduce the concept of 'information propagation' which induces the reward distribution shift between the two subsequent temporal models, and examine how it arises from the perspective of temporal models. To illustrate this, we consider

two temporal models $\mathcal{M}_t = \langle \mathbf{U}_t, \mathbf{V}_t, \mathcal{F}_t, P(\mathbf{U}_t \mid \mathbf{u}_t^\star)\rangle$ and $\mathcal{M}_{t+1} = \langle \mathbf{U}_{t+1}, \mathbf{V}_{t+1}, \mathcal{F}_{t+1}, P(\mathbf{U}_{t+1} \mid \mathbf{u}_{t+1}^\star)\rangle$. To isolate the effect of structural changes, we fix the exogenous distributions by assuming $P(\mathbf{U}_t \mid \mathbf{u}_t^\star) = P(\mathbf{U}_{t+1} \mid \mathbf{u}_{t+1}^\star)$ and $P(\mathbf{u}_t^\star) = P(\mathbf{u}_{t+1}^\star)$. The agent identifies that the data generation for $\mathcal{M}_{t+1}$ occurs only after the data generation governed by $\mathcal{M}_t$. Therefore, the agent at time t can neither observe the values of variables at a future time nor intervene in advance on future variables. For a fixed underlying SCM, the effective local functions can differ between time steps because their external-parent arguments differ. Time-dependent structural functions can also cause changes independently of information propagation; graph topology alone does not rule this out. Such function inequality arises due to predetermined values $\mathbf{v}_{t+1}^\star \in \mathscr{D}(\mathbf{V}_{t+1}^\star)$ where $\mathbf{V}_{t+1}^\star = \mathbf{Pa}_{V_{t+1}} \setminus \mathbf{V}_{t+1}$, which is determined at time step t. As a result, information propagation refers to the process by which the predetermined values, generated within a temporal model $\mathcal{M}_t$, influence the generation of a value in another temporal model $\mathcal{M}_{t+1}$.

To understand the concept better, we examine a simple case of two subsequent temporal models depicted in Fig. 2. At time step t, the value of X_t is generated from the function $f_{X_t} = U_{X_t}$ and the rest of the values are generated according to the topological (causal) order. At time step t', the function generating $X_{t'}$ is $f_{X_{t'}} = U_{X_{t'}} \oplus z_t$, where z_t is the value of Z_t generated at time step t. Thus, at time step t', the function $f_{X_{t'}}$ utilizes z_t, treating it as a conditional parameter (i.e., constant at t') to generate the value of $X_{t'}$. In this setting, the reward distribution of the arms may change over time, introducing additional complexity and challenges.

Graph representation of the NS-SCM-MAB The graph representation of the NS-SCM-MAB captures the dynamics of the underlying non-stationary structure and the dependencies between variables of different time slices. Each time slice corresponds to a set of variables $\mathbf{V}_t \in \mathbb{V}$ and the relationships between these slices are depicted by edges, which we elaborate on in the sequel, representing the information propagation across time. To establish the conditional independence structure between the causal diagrams of the NS-SCM-MAB and its probability distribution, we assume the *time-slice Markov property* as a temporal extension of the Markov property.

Assumption 3.1 (Time-slice Markov)**.** The temporal graph has directed transitions only from slice $t-1$ to slice t, and has no cross-slice unobserved confounding. Consequently, its induced distribution obeys $\mathbf{V}_t \perp\!\!\!\perp \mathbf{V}_{<t-1} \mid \mathbf{V}_{t-1}$ for $t \geq 3$. These structural restrictions, rather than a conditional-independence assertion alone, specify the first-order setting used here.

Assumption 3.1 illustrates that under non-stationary dynamics, the SCM-MAB exhibits a first-order Markovian structure where each time slice depends only on the immediately preceding one.

In addition, the graphical structure and topology of each time slice are identical across all time steps; that is, $\mathcal{G}[\mathbf{V}_t] = \mathcal{G}[\mathbf{V}_{t'}]$ for all $t \neq t' \in [T]$. This reflects the bandit nature of our setting, where each temporal model corresponds to a repeated instance of the same underlying causal structure. The equality identifies corresponding variables across slices. Given any two time slices $\mathcal{G}[\mathbf{V}_t]$ and $\mathcal{G}[\mathbf{V}_{t'}]$, this property implies stationarity between the corresponding temporal model MABs $\langle \mathcal{M}_t, Y_t\rangle$ and $\langle \mathcal{M}_{t'}, Y_{t'}\rangle$ provided that no information propagation occurs across time steps and the aligned structural functions and exogenous laws also agree.

If there exists at least one edge between $\mathcal{G}[\mathbf{V}_t]$ and $\mathcal{G}[\mathbf{V}_{t'}]$, where $V_t \in \mathbf{V}_t$ and $V_{t'} \in \mathbf{V}_{t'}$, this edge is represented as a *transition edge* $(V_t, V_{t'})$. The edge indicates that information propagation from a temporal model $\mathcal{M}_t$ to another $\mathcal{M}_{t'}$, conforming to the two subsequent time slices $\mathcal{G}[\mathbf{V}_t]$ and $\mathcal{G}[\mathbf{V}_{t'}]$, occurs at every time step. Notably, Assumption 3.1 implies that transition edges may exist only between consecutive time slices (i.e., first-order), thereby prohibiting connections between non-adjacent time slices. Paths through several consecutive slices can still transmit effects over an arbitrarily long horizon; first-order dependence does not justify limiting a planning search to two slices.

4 Motivating Example

The primary challenge we address is identifying action *sequences* that maximize the expected reward under the NS-SCM-MAB setting. We first contemplate the solution of the stationary SCM-MAB problem given a causal diagram $\mathcal{G}$ to understand the non-stationary causal bandit. In the stationary MAB setting, choosing a set of arms can be interpreted as identifying a set of nodes $\mathbf{Z}$ from $\mathbf{V} \setminus \{Y\}$

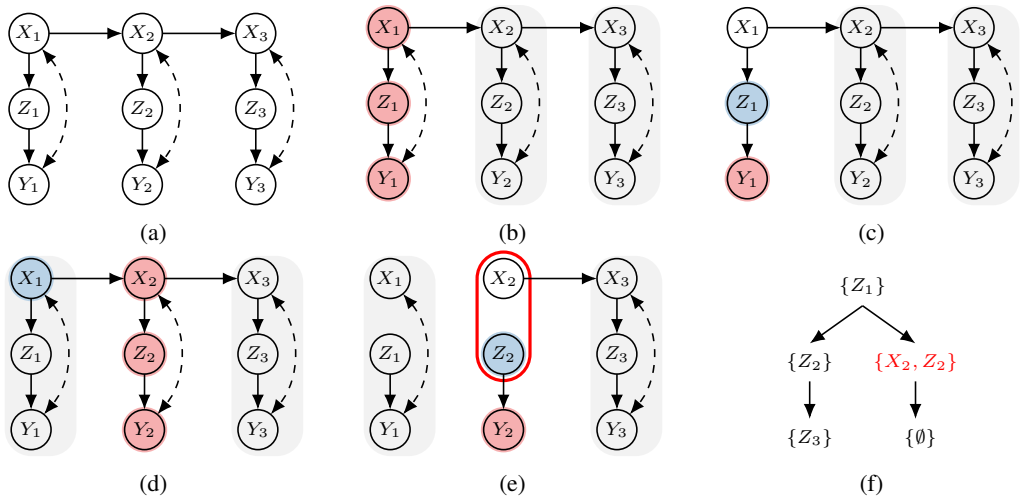


Figure 3: (a) Non-stationary causal diagram $\mathcal{G}$ with $t \in \{1, 2, 3\}$. (b-d) illustrate examples of MUCT/IB based on the agent's movement at each time step. (e) represents the agent's non-myopic intervention strategy at $t = 2$ (red box). (f) depicts a tree diagram of all possible scenarios when action selection starts with $\{Z_1\}$ (red text denotes POMIS$^+$).

that maximize the expected reward for outcome variable Y, formally written as

$$\mathbf{z}^\dagger \triangleq \underset{\mathbf{z} \in \mathscr{D}(\mathbf{Z}), \mathbf{Z} \subseteq \mathbf{V} \setminus \{Y\}}{\arg\max} \mathbb{E}[Y \mid \text{do}(\mathbf{Z} = \mathbf{z})]. \tag{1}$$

Previously, Lee and Bareinboim [2018] proposed a *do-calculus*-based approach to exclude intervention sets that may induce sub-optimal expected rewards, ultimately identifying possibly optimal minimal intervention sets (POMISs) [Lee and Bareinboim, 2018, Def. 2]. We demonstrate, however, that this approach may itself lead to sub-optimality in the NS-SCM-MAB setting.

In the non-stationary setting, unlike the stationary setting where the agent only computes a POMISs once, the agent explores the search space at every time step to identify the POMISs for each time-indexed reward variable. For instance, in Fig. 3(a), we may consider the process by which an agent selects an intervention set at each time step. We can identify the POMISs by utilizing the concepts of Minimal UC-Territory (MUCT) [Lee and Bareinboim, 2018, Def. 3] and Interventional Border (IB) [Lee and Bareinboim, 2018, Def. 4], which characterize the POMISs graphically.

At time step $t = 1$, from the agent's perspective, the set of possible exploration sets is $\{\emptyset, \{X_1\}, \{Z_1\}, \{X_1, Z_1\}\}$. We use an algorithm enumerating all POMISs [Lee and Bareinboim, 2018, Alg. 1] with respect to $[\![\mathcal{G}[\mathbf{V}_1], Y_1]\!]$. The algorithm generates $\text{IB}(\mathcal{G}[\mathbf{V}_1], Y_1) = \emptyset$ and $\text{IB}(\mathcal{G}[\mathbf{V}_1]_{\overline{X_1}}, Y_1) = \{Z_1\}$ from $\text{MUCT}(\mathcal{G}[\mathbf{V}_1], Y_1) = \{X_1, Z_1, Y_1\}$ and $\text{MUCT}(\mathcal{G}[\mathbf{V}_1]_{\overline{X_1}}, Y_1) = \{Y_1\}$. These are illustrated in Fig. 3(b) and Fig. 3(c), with the MUCT shaded in red and the IB shaded in blue. Thus, we get POMISs with respect to $[\![\mathcal{G}[\mathbf{V}_1], Y_1]\!]$—i.e., $\text{POMIS}_1 = \{\emptyset, \{Z_1\}\}$. As a result, following the same procedure at each time step $t = 1, 2, 3$, the agent obtains the following POMISs:

$$\text{POMISs}_1 = \{\emptyset, \{Z_1\}\}, \text{POMISs}_2 = \{\emptyset, \{Z_2\}\}, \text{POMISs}_3 = \{\emptyset, \{Z_3\}\}$$

Up to these time steps, the agent has identified the intervention set within its observable range at each time step. We next examine whether the identified intervention sets remain valid over an extended time horizon under the non-stationary setting. As illustrated with shaded blue in Fig. 3(d), $\text{MUCT}(\mathcal{G}[\mathbf{V}_1 \cup \mathbf{V}_2], Y_2)$ induces $\text{IB}(\mathcal{G}[\mathbf{V}_1 \cup \mathbf{V}_2], Y_2) = \{X_1\}$. Consequently, under the broader causal diagram $\mathcal{G}[\mathbf{V}_1 \cup \mathbf{V}_2]$, we deduce that $\text{POMIS}_2 = \{\{X_1\}, \{Z_2\}\}$. By applying the same procedure to $[\![\mathcal{G}[\mathbf{V}_1 \cup \mathbf{V}_2 \cup \mathbf{V}_3], Y_3]\!]$, we obtain $\text{POMIS}_3 = \{\{X_2\}, \{Z_3\}\}$. Eventually, we obtain the following POMISs with a broader horizon:

$$\text{POMISs}_1 = \{\emptyset, \{Z_1\}\}, \text{POMISs}_2 = \{\{X_1\}, \{Z_2\}\}, \text{POMISs}_3 = \{\{X_2\}, \{Z_3\}\}$$

This differs from the previously computed myopic evaluations. These discrepancies highlight that variables such as $\{X_1\}$ or $\{X_2\}$, excluded under myopic evaluations, may be optimal when viewed from a broader temporal perspective—while seemingly valid options like $\emptyset$ become sub-optimal.

Therefore, the agent should have intervened on $\{X_1\}$ and $\{X_2\}$ in advance because those variables are not manipulable for each current time step; $\{X_1\}$ for $t = 2$ and $\{X_2\}$ for $t = 3$. However, the variables may not have been considered optimal for a reward variable at an advanced time step. This is supported by the inequality below:

$$\max_{x_2 \in \mathscr{D}(X_2)} \mu_{x_2} \leq \max_{z_2 \in \mathscr{D}(Z_2)} \mu_{z_2} \tag{2}$$

which is derived from $\mu_{x_2^*} = \sum_{z_2} \mu_{z_2} P(z_2 | \operatorname{do}(x_2^*)) \leq \sum_{z_2} \mu_{z_2^*} P(z_2 | \operatorname{do}(x_2^*)) = \mu_{z_2^*}$ given $[\![\mathcal{G}[\mathbf{V}_2], Y_2]\!]$. This inequality shows that the expected reward for intervening on $\{X_2\}$ is weakly dominated by that of $\{Z_2\}$, so $\{X_2\}$ need not be retained as a possibly strictly optimal intervention set for Y_2. Nonetheless, this does not imply that $\{X_2\}$ should be excluded from the intervention set. Instead, it suggests that $\{X_2\}$ and $\{Z_2\}$ must be jointly intervened upon (i.e., $\{X_2, Z_2\}$ in Fig. 3(e)): while $\{Z_2\}$ is required to optimize the immediate reward Y_2, $\{X_2\}$ is necessary to influence the future reward Y_3. Importantly, intervening on $\{X_2\}$ does not block the causal influence of $\{Z_2\}$ on Y_2, thereby allowing both short-term and long-term rewards to be considered simultaneously. Similarly, for $t = 1$, the intervention set is assembled as $\{X_1, Z_1\}$. Finally, we obtain the following illustrative intervention sequences (without asserting an exhaustive characterization):

$$\begin{gathered}\{(\{Z_1\}, \{Z_2\}, \{Z_3\}), \quad (\{Z_1\}, \{X_2, Z_2\}, \emptyset), \\ (\emptyset, \{Z_2\}, \{Z_3\}), \quad (\{X_1, Z_1\}, \{Z_2\}, \{Z_3\}), \\ (\emptyset, \{X_2, Z_2\}, \emptyset), \quad (\{X_1, Z_1\}, \emptyset, \{Z_3\})\}.\end{gathered}$$

The details are formalized in Def. 5.1. We can depict two of these sequences as shown in Fig. 3(f), where each branch illustrates a valid path if Z_1 is intervened at the first time. These combinations will be discussed in detail in §5.

Building on the above example, we formally define the concept of POMIS for the NS-SCM-MAB and describe graphical candidates for future support, subject to the conditions and limitations in the following sections.

5 Conservation of Optimality

We examine theoretical guarantees for identifying optimal intervention sets under the NS-SCM-MAB problem. From §4, we know that myopic intervention (i.e., pulling arms within the current manipulable variables) does not always guarantee optimality for a broader horizon.

Rather, the agent must consider the sequence in which variables should be intervened on for subsequent time steps—a *non-myopic* strategy. To formalize this notion, we define an intervention sequence as follows.

Definition 5.1 (Intervention Sequence)**.** Given $[\![\mathcal{G}, \mathbf{Y}]\!]$, let $\mathbb{S} = (\mathbf{X}_1, \ldots, \mathbf{X}_T)$ be an intervention sequence where $\mathbf{X}_t \subseteq \mathbf{V}_t \setminus \{Y_t\}$ for $t \in [T]$.

An intervention sequence specifies the sets of variables available for intervention at each time step. It does not specify intervention values or a history-dependent policy. Values for an entire sequence may have to be chosen jointly to optimize cumulative reward. The following result concerns the best boundary context for a fixed future intervention set; it does not assert that every graph-level POMIS is optimal in an arbitrary SCM.

Theorem 5.1. *(Existence of Optimal Partial Assignment). Fix a future time t' and an intervention set $\mathbf{X}_{t'} \subseteq \mathbf{V}_{t'} \setminus \{Y_{t'}\}$. Let $\mathbf{D}_{t'} = Pa(\mathbf{V}_{t'}) \setminus \mathbf{V}_{t'}$ be the external-parent boundary. Fix any exogenous conditioning used in the temporal models, and let $\mathcal{H} \subseteq \mathscr{D}(\mathbf{D}_{t'})$ be a nonempty finite set of boundary assignments for which the temporal models are well-defined and the reward expectations are finite. Assume that $\mathscr{D}(\mathbf{X}_{t'})$ is nonempty and finite. For $\mathbf{v} \in \mathcal{H}$, define*

$$g_{\mathbf{X}_{t'}}(\mathbf{v}) = \max_{\mathbf{x} \in \mathscr{D}(\mathbf{X}_{t'})} \mathbb{E}^{\mathcal{M}_{t'}|\mathbf{v}}[Y_{t'} \mid do(\mathbf{X}_{t'} = \mathbf{x})], \qquad G_{\mathbf{X}_{t'}} = \max_{\mathbf{v} \in \mathcal{H}} g_{\mathbf{X}_{t'}}(\mathbf{v}).$$

There exist $\mathbf{W} \subseteq \mathbf{D}_{t'}$ and assignments $\mathbf{w}^*, \mathbf{x}^*$ such that the cylinder $\{\mathbf{v} \in \mathcal{H} : \mathbf{v}[\mathbf{W}] = \mathbf{w}^*\}$ is nonempty and

$$\mathbb{E}^{\mathcal{M}_{t'}|\mathbf{v}}[Y_{t'} \mid do(\mathbf{X}_{t'} = \mathbf{x}^*)] = G_{\mathbf{X}_{t'}}$$

throughout that cylinder. An inclusion-minimal such $\mathbf{W}$ exists.

The theorem maximizes over values of the specified intervention set and over boundary contexts. It does not compare that intervention set with other sets. A single future assignment is used throughout the asserted cylinder. A graph-level POMIS need not be optimal in the particular SCM under consideration. Neither $\mathbf{W} \cup \mathbf{X}_{t'}$ being an MIS of the expanded graph nor the existence of an admissible intervention on $\mathbf{W}$ follows from this attainment result.

Definition 5.2 (POMIS with Future Support (POMIS$^+$))**.** Fix $t < t'$, the future intervention set $\mathbf{X}_{t'}$, and the temporal-model family $\mathcal{H}$ in Thm. 5.1. Let $\mathbf{A}_t$ be a POMIS with respect to $[\![\mathcal{G}[\mathbf{V}_t], Y_t]\!]$, and set

$$\mathbf{C}_{t,t'} = (Pa(\mathbf{V}_{t'}) \setminus \mathbf{V}_{t'}) \cap (\mathbf{V}_t \setminus \{Y_t\}).$$

An admissible set $\mathbf{B}_t \subseteq \mathbf{C}_{t,t'}$ is future support for the specified family and intervention set if some assignments $\mathbf{b}_t^*, \mathbf{x}_{t'}^*$ give a nonempty cylinder in $\mathcal{H}$ and

$$\mathbb{E}^{\mathcal{M}_{t'}|\mathbf{v}}[Y_{t'} \mid do(\mathbf{X}_{t'} = \mathbf{x}_{t'}^*)] = G_{\mathbf{X}_{t'}} \quad \text{for every } \mathbf{v} \in \mathcal{H} \text{ such that } \mathbf{v}[\mathbf{B}_t] = \mathbf{b}_t^*.$$

If, in addition,

$$\mathbb{E}^{\mathcal{N}_t}[Y_t \mid do(\mathbf{A}_t = \mathbf{a})] = \mathbb{E}^{\mathcal{N}_t}[Y_t \mid do(\mathbf{A}_t = \mathbf{a}, \mathbf{B}_t = \mathbf{b})]$$

for every temporal model $\mathcal{N}_t$ conforming to $\mathcal{G}[\mathbf{V}_t]$ with finite reward expectations, every fixed temporal boundary context, and all assignments $\mathbf{a}, \mathbf{b}$ that agree on $\mathbf{A}_t \cap \mathbf{B}_t$, then $\mathbf{A}_t \cup \mathbf{B}_t$ is called a POMIS with specified future support. We denote it by POMIS$^+_{t,t'}$ relative to the specified future model family and $\mathbf{X}_{t'}$.

This definition makes the dependence on the future model and intervention set explicit. Future support is an additional condition; it is not certified by the graph-level POMIS property. The full boundary may include reward variables or variables at other times that cannot be intervened on at time t, so Thm. 5.1 does not guarantee that an admissible $\mathbf{B}_t$ exists. The definition also gives no guarantee that jointly chosen intervention values maximize cumulative reward.

Proposition 5.1. *(Admissible Time-t Interventions). Every POMIS$^+_{t,t'}$ in Def. 5.2 is contained in* $\mathbf{V}_t \setminus \{Y_t\}$.

This inclusion follows directly from the definition and does not require the time-slice Markov assumption. It does not establish that a two-slice window is sufficient for optimal planning over the full horizon.

6 Graphical Candidates and Conditional Composition

To construct graphical candidates for POMIS$^+$, we introduce two graphical concepts—interventional borders over future rewards (IB$^+$) and qualified interventional borders (QIB)—which, under the conditions below, enable the construction of POMIS$^+$. The graphical candidates do not automatically satisfy the future-support condition above.

Definition 6.1 (IB for the Subsequent Time Steps (IB$^+$))**.** Given $[\![\mathcal{G}, \mathbf{Y}]\!]$, for $t < t'$ and any $\mathbf{W} \subseteq An(Y_{t'})_{\mathcal{G}[\bigcup_{i=t}^{t'} \mathbf{V}_i]} \setminus \{Y_{t'}\}$, define $\mathbf{Z}(\mathbf{W}) = \text{IB}(\mathcal{G}[\bigcup_{i=t}^{t'} \mathbf{V}_i]_{\overline{\mathbf{W}}}, Y_{t'})$. Then, for each $j \in \{t, \ldots, t'\}$, $\mathbf{Z}(\mathbf{W}) \cap \mathbf{V}_j$ is called an Interventional Border at time slice j for $Y_{t'}$, denoted by $\text{IB}^+_{j,t'}(\mathbf{W})$. A raw candidate need not be an admissible action set. In particular, a candidate containing Y_j is not available for intervention at time j. Only candidates contained in $\mathbf{C}_{t,t'}$ can be considered as support in Def. 5.2.

Intuitively, $\text{IB}^+_{t,t'}$ is computed for the future reward $Y_{t'}$ but resides within $\mathbf{V}_t$. Therefore, we must identify a *qualified* interventional border under $[\![\mathcal{G}[\mathbf{V}_t], Y_t]\!]$—that is, an IB that satisfies certain conditions to remain effective even when $\text{IB}^+_{t,t'}$ is fixed.

Definition 6.2 (Qualified Immediate-Reward IB)**.** Let $H_t = \mathcal{G}[\mathbf{V}_t]$ and let $\mathbf{B} \subseteq \mathbf{V}_t \setminus \{Y_t\}$. A set $\mathbf{A}$ belongs to $QIB_t(\mathbf{B})$ if all of the following hold:

1. $\mathbf{A}$ is a POMIS with respect to $[\![H_t, Y_t]\!]$;
2. for some $\mathbf{L} \subseteq \mathbf{V}_t \setminus \{Y_t\}$, $IB((H_t)_{\overline{\mathbf{B} \cup \mathbf{L}}}, Y_t) = \mathbf{A}$;
3. $(\mathbf{B} \setminus \mathbf{A}) \cap An(Y_t)_{(H_t)_{\overline{\mathbf{A}}}} = \emptyset$.

Thus $QIB_t(\mathbf{B})$ is a family of intervention sets. The final condition states that, after intervening on $\mathbf{A}$, no directed path from $\mathbf{B} \setminus \mathbf{A}$ to Y_t remains.

Proposition 6.1. *(Conditional Composition). Fix the future temporal-model family and intervention set in Def. 5.2. Suppose that an admissible candidate $\mathbf{B} \subseteq \mathbf{C}_{t,t'}$ satisfies that definition's future-support condition, and that $\mathbf{A} \in QIB_t(\mathbf{B})$. Then $\mathbf{A} \cup \mathbf{B}$ is a POMIS$^+_{t,t'}$ relative to those specified choices.*

The future-support premise must be checked separately. A time restriction of an IB can fail it, even when some assignment to the full boundary makes the chosen future-slice POMIS optimal; a counterexample is given in Appendix G. The unconditional composition statement and Proposition G.1 of the published version are withdrawn. These restricted statements do not establish the published algorithm's soundness or completeness, or the sufficiency of a two-slice window for cumulative-reward optimization.

As shown in Fig. 4, an IB for $Y_{t'}$ spans over two time steps (blue shaded), and $\{X_t\}$ is an IB$^+_{t,t'}$ from Def. 6.1. The candidate $\mathbf{B} = \{X_t\}$ and $\mathbf{A} = \{Z_t\}$ satisfy the immediate-reward condition: intervening on Z_t blocks every directed path from X_t to Y_t. This verifies the displayed immediate-reward equality. It does not, by itself, verify optimal future values for a specified SCM or optimality of the full intervention sequence.

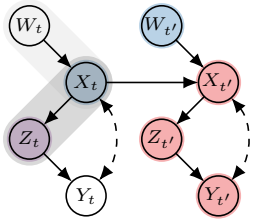


Figure 4: POMIS for the subsequent time step.

7 An Optimal-Value-Preserving Static Reduction

For a complete intervention sequence fixed before the episode, the cumulative-reward problem has a direct reduction to the static SCM-MAB setting with non-manipulable variables [Lee and Bareinboim, 2018, 2019]. This supplies a general optimal-value guarantee without the local future-support construction. It is an application of the existing static reduction, rather than an additional temporal characterization.

Let the known finite acyclic time-expanded graph be $\mathcal{G}$, with slices $\mathbf{V}_1, \ldots, \mathbf{V}_T$ partitioning $\mathbf{V}$ and reward set $\mathbf{Y} = \{Y_1, \ldots, Y_T\}$. Let $\mathbf{A} \subseteq \mathbf{V} \setminus \mathbf{Y}$ be the allowed intervention variables; ordinarily $\mathbf{A} = \mathbf{V} \setminus \mathbf{Y}$. An allowed fixed plan chooses any $\mathbf{S} \subseteq \mathbf{A}$ and any assignment $\mathbf{s} \in \mathscr{D}(\mathbf{S})$. Its chronological representation is $(\mathbf{S} \cap \mathbf{V}_1, \ldots, \mathbf{S} \cap \mathbf{V}_T)$ with the corresponding assigned values. The plan does not depend on realized observations. We assume nonempty finite domains for every variable in $\mathbf{A}$, permit every such subset and assignment, and impose no additional coupled budgets or individual-reward constraints. Here $\mathscr{D}(V)$ is the full endogenous state space of V, and every value in it is available for intervention; it is not a restricted set of allowed control values. The empty target has the single empty assignment.

Add a new non-manipulable sink $R = \sum_{t=1}^{T} Y_t$, with an incoming edge from every Y_t, and call the augmented graph $\mathcal{G}^R$. Every original reward remains non-manipulable. Project $\mathcal{G}^R$ onto $\mathbf{A} \cup \{R\}$, removing every other endogenous variable, and write $H = \mathrm{LP}_{\mathbf{A} \cup \{R\}}(\mathcal{G}^R)$. The graph H retains action time labels and accounts for induced common causes. For the proof, augment any compatible SCM and substitute the removed equations to obtain a reduced SCM $\mathcal{M}_H$, preserving the original exogenous vector and joint law. The latent projection H is compatible with this reduced SCM. Algorithm 1 needs only the graph and allowed variables; it does not require the unknown structural equations or noise law.

Using the ancestor-restricted territory and border of Defs. B.4 and B.5, with incoming directed and incident bidirected edges removed by surgery, define the raw static-border family

$$\mathcal{C}(H, R) = \{\mathrm{IB}(H_{\overline{\mathbf{W}}}, R) : \mathbf{W} \subseteq \mathbf{A}\}.$$

The ordinary static POMIS algorithm can enumerate this same graphical family. The explicit enumeration below deduplicates target supports and avoids relying on an unproved temporal pruning rule. Each returned support denotes all its fixed value assignments.

Theorem 7.1 (Optimal-value coverage for fixed plans)**.** *Suppose the SCM is acyclic and compatible with the known full graph, using an expanded DAG with independent exogenous roots and all shared*

Algorithm 1 Static-border candidates for fixed intervention plans

function AggregateCandidates($\mathcal{G}, \mathbb{V}, \mathbf{Y}, \mathbf{A}$)
Add the non-manipulable sink $R = \sum_{t=1}^{T} Y_t$ to obtain $\mathcal{G}^R$
$H \leftarrow \mathrm{LP}_{\mathbf{A}\cup\{R\}}(\mathcal{G}^R)$; $\mathbb{S} \leftarrow \emptyset$
for each $\mathbf{W} \subseteq \mathbf{A}$ **do**
  $\mathbf{B} \leftarrow \mathrm{IB}(H_{\overline{\mathbf{W}}}, R)$
  $\mathbb{S} \leftarrow \mathbb{S} \cup \{(\mathbf{B} \cap \mathbf{V}_1, \ldots, \mathbf{B} \cap \mathbf{V}_T)\}$
return $\mathbb{S}$, retaining every assignment in $\mathscr{D}(\mathbf{B})$ for each returned target set $\mathbf{B}$

uses represented in the graph, and $\mathbb{E}[|Y_t| \mid do(\mathbf{s})] < \infty$ *for every allowed plan and every* t*. Under the finite-domain and action assumptions above, Alg. 1 terminates, returns a nonempty family of legal fixed plans, and satisfies*

$$\max_{\substack{\mathbf{B}\in\mathcal{C}(H,R)\\ \mathbf{b}\in\mathscr{D}(\mathbf{B})}} \mathbb{E}^{\mathcal{M}}\left[\sum_{t=1}^{T} Y_t \mid do(\mathbf{b})\right] = \max_{\substack{\mathbf{S}\subseteq\mathbf{A}\\ \mathbf{s}\in\mathscr{D}(\mathbf{S})}} \mathbb{E}^{\mathcal{M}}\left[\sum_{t=1}^{T} Y_t \mid do(\mathbf{s})\right].$$

This holds for each fixed compatible SCM and its given action domains. It permits arbitrary forward temporal edges, reward variables that are parents of later action variables, and cross-slice confounding. Repeated slice topology and first-order transitions are unnecessary.

Proof. Forward substitution preserves the aggregate reward under each retained intervention. In the resulting static model, each intervention mean is a probability-weighted average of intervention means on its associated border $\mathrm{IB}(H_{\overline{\mathbf{S}}}, R)$, and hence cannot exceed the best assignment to that border. This border is enumerated by the algorithm. The reverse inequality follows because every enumerated assignment is allowed. The exact identity and its mixing weights appear in Lem. H.1. □

What this recovers. The theorem recovers global optimal-value coverage for complete fixed sequences. It does not assert that these sequences are exactly the POMIS$^+$ objects in Def. 5.2, or that every returned border has a strict-optimality witness within the subclass whose aggregate equation is the prescribed sum. Static graph-level POMIS enumeration is exact for its ordinary graph-only model class with freely chosen witness domains; imposing the sum equation can make that family conservative. Nor does value coverage establish optimality among policies depending on observed history, feasibility under additional constraints, or a finite-horizon learning advantage.

Why the published future-border restriction can fail. Let independent fair binary sources satisfy $X_t = U_t$, and set $Y_1 = \frac{3}{4}\mathbf{1}\{X_1 = U_1\}$ and $Y_2 = \frac{3}{4}\mathbf{1}\{X_2 = U_2\} + \frac{1}{4}X_1$. Each slice contains $X_t \to Y_t$ and $X_t \leftrightarrow Y_t$, and the sole transition is $X_1 \to Y_2$. The action variables are $\mathbf{A} = \{X_1, X_2\}$ with full binary domains. Every border for Y_2 includes X_1: the two borders are $\{X_1\}$ and $\{X_1, X_2\}$. Thus the intended backward construction, even with its tuple order corrected, restricts the best expected sum to $11/8$. The empty intervention plan gives $E[Y_1] = 3/4$ and $E[Y_2] = 7/8$, totaling $13/8$. Fixing $X_1 = 1$ instead gives $3/8 + 1 = 11/8$: the future gain $1/8$ costs $3/8$ in the earlier reward. By contrast, projecting onto $\{X_1, X_2, R\}$ gives $X_1 \to R$, $X_2 \to R$, $X_1 \leftrightarrow R$, and $X_2 \leftrightarrow R$. Its uncut border is empty, so Alg. 1 retains this global maximizer. The value gap refutes the claimed cumulative-reward justification of the local restriction; it does not identify an exact POMIS$^+$ sequence class.

Proposition 7.1 (Finite construction cost)**.** *For an explicitly given acyclic mixed graph with* $m = |\mathbf{V}| + 1$ *vertices, latent projection and each territory/border computation can be performed in polynomial time in* m*. The raw enumeration in Alg. 1 has* $2^{|\mathbf{A}|}$ *iterations and takes* $\mathcal{O}(\mathrm{poly}(m, T)2^{|\mathbf{A}|})$ *time with polynomial set operations and efficient deduplication. It returns at most* $2^{|\mathbf{A}|}$ *distinct target sets. This cost concerns target sets; explicitly listing all their value assignments incurs the additional size* $\sum_{\mathbf{B}\in\mathcal{C}(H,R)} |\mathscr{D}(\mathbf{B})|$.

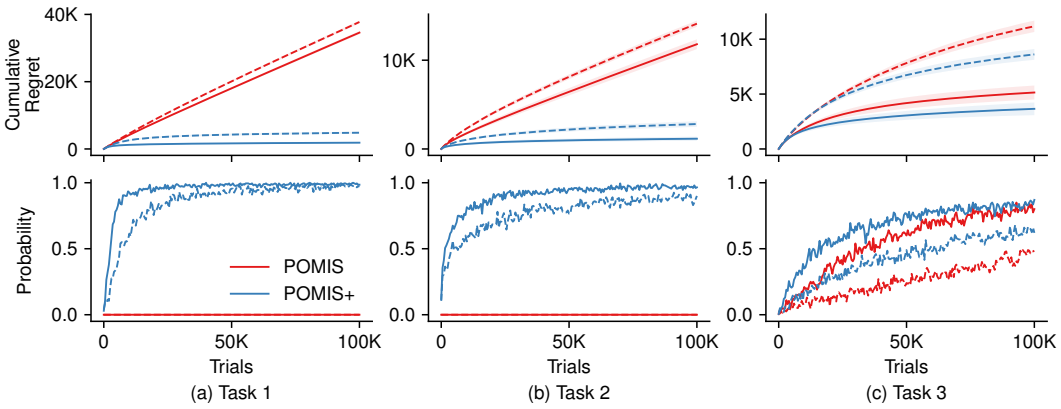


Figure 5: Cumulative regret (top) and optimal arm selection probabilities (bottom) across tasks (columns), with solid for Thompson Sampling and dashed for KL-UCB; CR shading denotes ± 1 SD. Published curves for the original temporal method are retained pending reproduction; see the correction note in §8.

8 Experiments

We evaluate how effectively the POMIS$^+$ strategy captures long-term effects compared to the myopic POMIS baseline. We conduct experiments[3] on three settings, each designed to highlight different aspects of temporal intervention planning. Detailed specifications for each task are provided in App. I. We report two metrics—cumulative regret (CR) and optimal arm selection probability (OAP)—under two MAB solvers: Thompson Sampling (TS) [Thompson, 1933] and KL-UCB [Cappé et al., 2013].

Each arm specifies a complete, fixed intervention sequence. The released implementation computes its expected total reward $\mu(a) = \mathbb{E}[\sum_t Y_t \mid \mathrm{do}(a)]$, then supplies Bernoulli feedback with success probability $\mu(a)/3$ to the MAB solver, including the two-step Task 2. CR is the pseudo-regret $\sum_{n=1}^{N}[\max_a \mu(a) - \mu(a_n)]$ in unscaled total-reward units; OAP is the across-repetition probability of selecting an arm attaining $\max_a \mu(a)$ on a given trial. The released reporting code uses approximate 95% confidence intervals for the $\pm$ values below; the CR shading in Fig. 5 shows ± 1 SD.

Correction note. The results below concern the original temporal candidate construction, not the aggregate static replacement in Alg. 1; learning experiments for that replacement have not been rerun. The original numerical values and curves are retained. The released multi-task runner retains candidate sequences between tasks, which can change the Task 1 arm list and its learning curves. The exact invocation and result arrays underlying the published curves have not been established, so their reproduction after correcting this state handling remains pending.

Task 1 (Fig. 3(a)): As shown in Fig. 5(a), CR of the TS for the POMIS$^+$ strategy converges around step 20K (solid blue line). In contrast, the sequences constructed from the myopic POMIS strategy show no sign of convergence (solid red line). Similar patterns are observed for the OAP result. Quantitatively, by step 100K, the POMIS$^+$ strategy achieves a CR of 1.9K$\pm$29 (TS) and 4.8K$\pm$43 (KL-UCB), with optimal-arm probabilities (OAP) of 98.5$\pm$1.7% and 97.0$\pm$2.4%, respectively. Meanwhile, the POMIS baseline performs substantially worse, with CR values exceeding 34K and 37K, while OAP scores remain near zero. This result is attributed to the fact that myopic strategies fail to include earlier intervention nodes such as X_1 for Y_2 and X_2 for Y_3. The inability to access such variables leads to critical long-term failures in non-stationary settings.

Task 2 (Fig. 4): As shown in the second column of Fig. 5, the POMIS$^+$ strategy consistently outperforms the myopic POMIS baseline in both cumulative regret and optimal arm selection probability. By step 100K, POMIS$^+$ achieves substantially lower regrets—1.1K$\pm$32 (TS) and 2.8K$\pm$44 (KL-UCB)—compared to 11.8K$\pm$77 and 14.1K$\pm$48 for the POMIS strategy. Furthermore, the OAP of POMIS$^+$ reaches 96.5$\pm$2.6% (TS) and 90.0$\pm$4.2% (KL-UCB), while the POMIS baseline remains near zero. These results demonstrate that POMIS$^+$ successfully captures the long-term effect of X_t on $Y_{t'}$, which is overlooked by the myopic strategy.

[3] A Python implementation can be found at: `https://github.com/yeahoon-k/NS-SCMMAB`.

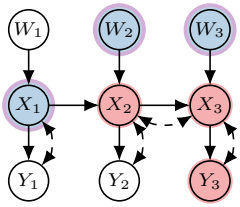


Figure 6: $\mathcal{G}$ for Task 3.

Task 3 (Fig. 6): The final experiment is based on the SCM, which intentionally violates the time-slice Markov assumption (Assumption 3.1) by introducing long-range dependencies and latent interactions across time (§F). As shown in Fig. 5(c), POMIS$^+$ outperforms the baseline, though the gap is smaller than in previous tasks. By step 100K, the CR of POMIS$^+$ is 3.7K$\pm$80 (TS) and 8.6K$\pm$69 (KL-UCB), improving upon the POMIS baseline regret of 5.1K$\pm$89 and 11.2K$\pm$72, respectively. In terms of OAP, POMIS$^+$ achieves 87.0$\pm$4.7% (TS) and 62.0$\pm$6.7% (KL-UCB), while POMIS reaches only 80.0$\pm$5.6% and 48.5$\pm$6.9%, respectively. For the specified SCM, both candidate families contain a globally optimal intervention sequence. POMIS$^+$ provides fewer instantiated arms (48 versus 64), so this example compares learning over different candidate families rather than demonstrating exclusion of the optimum by the myopic family.

9 Discussion

We assume access to the full causal structure during evaluation. In many real-world settings, however, the agent may have only limited or partial access to future causal diagrams rather than complete knowledge of them. Extending our framework to handle partially observed or uncertain temporal structures—while maintaining optimality—remains an important direction for future work.

While the aggregate static reduction preserves the optimal value of complete fixed plans, its computational complexity is constrained by the exponential growth of the search space. A promising direction is to exploit repeated or modular structures that frequently appear in non-stationary environments—for instance, by developing segment-wise variants of our algorithm that reuse computed intervention sequences across recurring causal patterns. Such extensions could preserve theoretical soundness while substantially improving scalability and applicability to larger, real-world domains.

For the experimental setting, each experimental trial selects a complete fixed intervention sequence and uses Bernoulli feedback with success probability $\mu(a)/3$, where $\mu(a)$ is its expected total reward. This episodic formulation enables fair comparison across intervention sequences based on their overall causal influence over time. Although interventions are not executed sequentially across evolving environments, the causal dependencies among time-indexed variables are fully encoded in the temporal models. The surrogate preserves expected-reward rankings but differs from sampled trajectory-sum feedback and does not establish general candidate coverage. Learning-curve reproduction remains pending (§8). Further discussions are provided in §K.

10 Conclusion

In this work, we present a novel framework; the NS-SCM-MAB, that integrates non-stationary multi-armed bandit problems with structural causal models. Our formulation captures both evolving reward distributions and shifting causal structures, enabling temporally-aware decision-making that accounts for how interventions propagate over time. To formalize this, we introduce POMIS$^+$, a temporally-extended notion of optimal intervention sets, and provide a fixed-set best-context result under finite domains. The original future-border restriction can lose a cumulative optimum. The aggregate static reduction applies existing SCM-MAB results to preserve the optimal value of a complete fixed plan under the conditions in §7; exact POMIS$^+$ characterization and adaptive-policy optimality are not established. The reported experiments concern the original construction; the replacement has not been evaluated by rerunning them. We believe this work provides a foundation for causal reasoning in sequential decision-making under temporal dynamics and opens up new avenues for research at the intersection of causal inference and non-stationary environments.

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A Organization

A schematic of our framework and its logical flow is illustrated in Fig. 7. The diagram visually maps out how the paper develops, starting from the formulation of the non-stationary SCM-MAB problem in §3, and introducing the temporal model along with key graphical assumptions about the nature of non-stationarity. §5 establishes that in a temporal model, partial assignments of predetermined variables can affect optimal interventions at future time steps, motivating the definition of POMIS$^+$ as a non-myopic intervention. This leads into the graphical characterization in §6, which defines key structures (IB$^+$, QIB) that construct POMIS$^+$ under the stated local conditions. Finally, §7 presents an algorithm that preserves the best fixed-plan value by aggregate static reduction. The structure clarifies the dependencies between the paper's components and highlights how our contributions build upon one another.

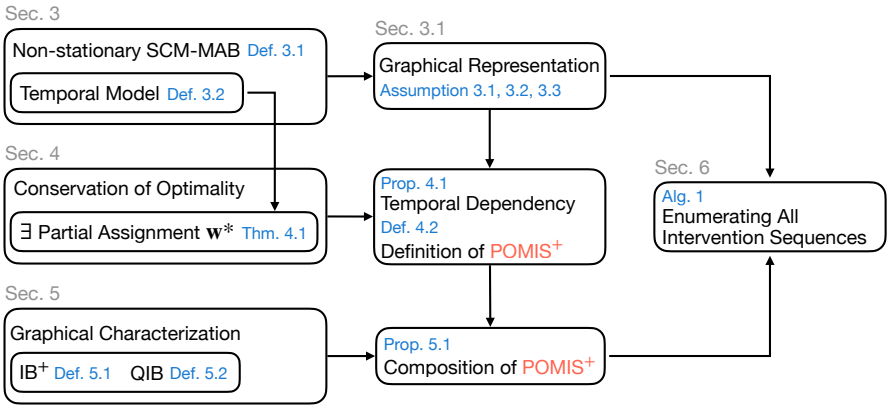


Figure 7: An overall schema of this paper.

B Preliminaries

Here we present definitions of MIS, POMIS, MUCT, and IB.

Definition B.1. (Minimal Intervention Set (MIS) [Lee and Bareinboim, 2018]). A set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is said to be a *minimal intervention set* relative to $[\mathcal{G}, Y]$ if there is no $\mathbf{X}' \subset \mathbf{X}$ such that $\mu_{\mathbf{x}[\mathbf{X}']} = \mu_{\mathbf{x}}$ for every SCM conforming to the $\mathcal{G}$.

Definition B.2. (Possibly-Optimal Minimal Intervention Set (POMIS) [Lee and Bareinboim, 2018]). Given information $[\mathcal{G}, Y]$, let $\mathbf{X}$ be a MIS. If there exists an SCM conforming to $\mathcal{G}$ such that $\mu_{\mathbf{x}^*} > \forall_{\mathbf{Z} \in \mathbb{Z} \setminus \{\mathbf{X}\}} \mu_{\mathbf{z}^*}$, where $\mathbb{Z}$ is the set of MISs with respect to $\mathcal{G}$ and Y, then $\mathbf{X}$ is a *possibly-optimal minimal intervention set* with respect to the information $[\mathcal{G}, Y]$.

Definition B.3. (C-component [Tian and Pearl, 2003]). In a causal diagram $\mathcal{G}$, two variables are said to be in the same confounded component (for short, C-component or $\mathsf{CC}(\cdot)_{\mathcal{G}}$) if and only if they are connected by a bi-directed edge (i.e., a path composed solely of $V_i \leftrightarrow V_j$).

In this paper, we denote $\mathsf{CC}(\mathbf{X})_{\mathcal{G}} = \bigcup_{X \in \mathbf{X}} \mathsf{CC}(X)_{\mathcal{G}}$.

Definition B.4. (Unobserved-Confounders' Territory [Lee and Bareinboim, 2018]). Given information $[\mathcal{G}, Y]$, let H be $\mathcal{G}[An(Y)_{\mathcal{G}}]$. A set of variables $\mathbf{T} \subseteq \mathbf{V}(H)$ containing Y, where $\mathbf{V}(H)$ is the set of variables in H, is called a *UC-territory* on $\mathcal{G}$ with respect to Y if $\mathrm{De}(\mathbf{T})_H = \mathbf{T}$ and $\mathsf{CC}(\mathbf{T})_H = \mathbf{T}$.

A UC-territory $\mathbf{T}$ is said to be *minimal* if no $\mathbf{T}' \subset \mathbf{T}$ is a UC-territory. A minimal UC-Territory (MUCT) for $\mathcal{G}$ and Y can be constructed by extending a set of variables, starting from $\{Y\}$, alternatively updating the set with the c-component and descendants of the set.

Definition B.5. (Interventional Border [Lee and Bareinboim, 2018]). Let $\mathbf{T}$ be a minimal UC-territory on $\mathcal{G}$ with respect to Y. Then, $\mathbf{X} = pa(\mathbf{T})_{\mathcal{G}} \setminus \mathbf{T}$ is called an *interventional border* for $\mathcal{G}$ with respect to Y.

Algorithm 2 Minimal unobserved confounders' territory

function $\mathsf{MUCT}(\mathcal{G}, \mathbb{Y})$
$H = \mathcal{G}[An(Y)_{\mathcal{G}}]$
$\mathbf{Q} = \{Y\}$; $\mathbf{T} = \{Y\}$
while $\mathbf{Q} \neq \emptyset$ **do**
  remove an element Q_1 from $\mathbf{Q}$
  $\mathbf{W} = \mathsf{CC}(Q_1)_H$; $\mathbf{T} = \mathbf{T} \cup \mathbf{W}$; $\mathbf{Q} = (\mathbf{Q} \cup de(\mathbf{W})_H) \setminus \mathbf{T}$
return $\mathbf{T}$

C Nomenclature

NS	Non-stationary
SCM	Structural causal model
RL	Reinforcement learning
MAB	Multi-armed bandit
MIS	Minimal intervention set
UC	Unobserved confounder
POMIS	Possibly-optimal minimal intervention set
MUCT	Minimal unobserved confounders' territory
IB	Interventional border
QIB	Qualified interventional border
DUC	Dynamically unobserved confounder
CR	Cumulative regret
OAP	Optimal arm-selection probability
KL-UCB	Kullback-Leibler upper confidence bound

D Stationary SCM-MAB

The cumulative regret of the stationary setting is given by:

$$R_T \triangleq \sum_{t=1}^{T} \Big(\mu_{\mathbf{x}^\dagger} - \mathbb{E}\big[\mu_{\mathbf{x}_t}\big] \Big), \tag{3}$$

where $\mathbf{x}^\dagger$ denotes the globally optimal arm, defined as

$$\mathbf{x}^\dagger \triangleq \underset{\mathbf{x} \in \mathscr{D}(\mathbf{X}),\, \mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}}{\arg\max} \mathbb{E}[Y \mid \mathrm{do}(\mathbf{X} = \mathbf{x})].$$

where $\mu_{\mathbf{x}_t}$ is the arm played at round t. The SCM-MAB setting assumes that the agent has full access to the causal graph $\mathcal{G}$ of $\mathcal{M}$, although its parametrization remains unknown—i.e., the agent knows the structure $\mathcal{G}$, but not the structural functions $\mathcal{F}$ or the distribution over exogenous variables $P(\mathbf{U})$. Furthermore, the causal graph $\mathcal{G}$ is assumed to be static, meaning that the underlying causal structure of the domain does not change over time. As a result, the agent interacts with the same causal model in each round.

In this setting, there is no confounding across time slices, and thus, no information propagates between rounds. Consequently, the reward distribution remains fixed across rounds.

In practice, under the stationary setting, the agent effectively observes, in each round, a causal diagram $\mathcal{G}$ composed of temporally disconnected slices, as illustrated in Fig. 1(b). This stationary (and also non-stationary) formulation typically assumes that the number of arms is constant throughout the interaction. The graphical structure and topology of each time slice are identical across all rounds.

The SCM-MAB framework inherently introduces dependencies between arms, stemming from the underlying causal relationships among endogenous and exogenous variables. Lee and Bareinboim [2018, 2019] identified two structural properties that can be derived from any SCM-MAB framework:

1. Arm equivalence: a characterization of arms that share identical reward distributions, determined using constraints from do-calculus, and
2. Partial-orders among arms: under what topological conditions one arm can be optimal.

Leveraging these properties, one can identify minimal intervention sets (MIS) that constitute a non-redundant collection of informative interventions. In addition, Lee and Bareinboim [2019] identified both MIS and POMIS for the stationary setting with non-manipulative variables. However, these characterizations rest on the assumption that the causal graph $\mathcal{G}$ remains static and stationary—meaning that no information is carried over from previous decisions. §3 extends beyond the stationary assumption to present the SCM-MAB framework in the non-stationary setting.

E Comparison with Conventional Non-Stationary Bandit Algorithms

We contrast our approach to non-stationarity in the SCM-MAB with traditional non-stationary bandit settings. Our focus is on how causal modeling provides a structured explanation for reward distribution shifts over time, as opposed to treating them as statistical artifacts. These build upon early foundational work on dynamic allocation and index policies [Whittle, 1988, Gittins, 1979].

Conventional NS-bandit formulations Conventional non-stationary bandit algorithms aim to maintain *low regret* with respect to a comparator (or competitor) class—a predefined set of benchmark policies that may adapt over time to account for changing environments (e.g., policies that allow a limited number of switches between arms, comparators that track the best-performing arm over recent time windows, or strategies that assume bounded changes in the underlying reward distribution). These algorithms are typically categorized into two regimes: adversarial and stochastic bandits, illustrated in Table 1.

Table 1: Representative non-stationary bandit settings and algorithms (adapted from Lattimore and Szepesvári [2020]).

Regime	**Setting**	**Description**	**Representative Algorithms**
Adversarial	L-switching	The identity of the optimal arm may change abruptly up to L times	Exp3.S, AdaHedge
Adversarial	Variation budget	Total variation in reward sequences is bounded by V	Rexp3, Adapt-EvE
Stochastic	Piecewise-stationary	Rewards are stationary within intervals, with occasional change-points	Sliding-window UCB, Change-point Thompson Sampling
Stochastic	Drifting	Expected rewards evolve smoothly over time	Discounted UCB, Sliding-window UCB, SW-TS

Each setting assumes non-stationarity as a statistical property: either abrupt shifts, or slowly drifting rewards. Some algorithms require knowledge of the number of switches L, while others are designed to be adaptive. In the variation budget setting, the cumulative amount of change in reward distributions is bounded by a budget V, offering a finer-grained control of non-stationarity than simple change-point models.

Causal non-stationarity (our perspective) Our framework models non-stationarity as a consequence of causal information propagation across time. Specifically, transition edges in the causal graph $\mathcal{G}$ (e.g., $X_t \to Z_{t'}$) induce changes in the downstream reward variables (e.g. $Y_{t'}$), illustrated in Fig. 2. This structure directly captures *why* the reward distribution changes.

For instance, whereas traditional settings might treat $\mathbb{E}[Y_t \mid \mathrm{do}(x)]$ as shifting arbitrarily with t, our approach identifies structural causes: $P(Y_t \mid \mathrm{do}(X_t))$ is influenced by information propagation

from previous variables (e.g. X_{t-1}, Z_{t-1}). This allows us to model the *mechanism* behind reward distribution shifts.

Moreover, the presence of arcs (or edges) between time slices in $\mathcal{G}$ determines where and how changes occur. This is closely aligned with the "mean payoff drift" interpretation in traditional models [Lattimore and Szepesvári, 2020, Chapter 31], but we interpret it in terms of explicit graphical information in $\mathcal{G}$.

Summary Most traditional algorithms detect or adapt to change, but do not explain it. Our approach, by contrast, offers a mechanism-based explanation of non-stationarity—one grounded in a causal understanding of the system under investigation, which is itself a strong assumption—via the SCM structure. This allows for:

- Identification of reward-relevant intervention targets (POMIS$^+$)
- Structural candidates for intervention planning, subject to the corrected local conditions and global limitations

Ultimately, our method treats non-stationarity as a structured phenomenon emergent from a dynamic causal model, rather than as an arbitrary change in observed statistics.

F Semi Time-Slice Markovian Non-Stationary SCM-MAB

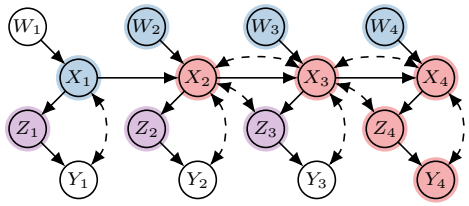


Figure 8: IB$^+$ ranges over four time steps.

We begin by formalizing dynamic unobserved confounders (DUCs), which represent exogenous variables inducing dependencies across time.

Definition F.1. (Dynamic unobserved confounders (DUCs)). Given $[\![\mathcal{G}, \mathbf{Y}]\!]$, let $\mathbf{U}_t^\star$ denote the set of exogenous variables that induce dependencies between variables in $\mathbf{V}_t$ and $\mathbf{V}_{t'}$ for some $t < t'$. If $U_j \in \mathbf{U}_t^\star$, then we refer to U_j as a dynamic unobserved confounder (DUC), indicating that U_j introduces confounding effects that persist across time.

When such DUCs are permitted in the graphical structure, we say the temporal graph satisfies the semi-time-slice Markov property.

Definition F.2 (Semi-time-slice Markov)**.** A temporal graph $\mathcal{G}$ satisfies the semi-time-slice Markov property if it allows the presence of dynamic unobserved confounders (DUCs) across time. That is, the graph permits bidirected edges between variables $V_t \in \mathbf{V}_t$ and $V_{t'} \in \mathbf{V}_{t'}$ for $t \neq t'$, representing confounding induced by exogenous variables $U_j \in \mathbf{U}_t^\star$ that simultaneously affect both time slices.

Under the semi-time-slice Markov assumption induced by dynamic unobserved confounders (Def. F.1), we need to perform a time-expanded search for valid intervention targets across multiple time steps. Consider the example in Fig. 8. When we calculate IB$^+$ for MUCT$(\mathcal{G}[\bigcup_{i=1}^4 \mathbf{V}_i], Y_4)$, the result consists of four sets: IB$^+_{1,4} = \{X_1\}$, IB$^+_{2,4} = \{W_2\}$, IB$^+_{3,4} = \{W_3\}$ and IB$^+_{4,4} = \{W_4\}$ (blue shaded). Given those IB$^+$ at each time step, we can determine each QIB$_t$ according to Def. 6.2. For example, when IB$^+_{1,4} = \{X_1\}$ is given, the IB corresponding to the MUCT$(\mathcal{G}[\mathbf{V}_1]_{\overline{X_1 \cup Z_1}}, Y_1) = \{Y_1\}$ is Z_1, hence QIB$_1 = \{Z_1\}$. Similarly, for MUCT$(\mathcal{G}[\mathbf{V}_2]_{\overline{X_2 \cup W_2}}, Y_2)$, we can select QIB$_2 = \{Z_2\}$, and for MUCT$(\mathcal{G}[\mathbf{V}_3]_{\overline{X_3 \cup W_3}}, Y_3)$, we can choose QIB$_3 = \{Z_3\}$ (purple shaded). Subject to the additional premises of the corrected Prop. 6.1, we can obtain POMIS$^+$s by taking the union of each IB$^+_{t,t'}$ and QIB$_t$—POMIS$^+_{1,4} = \{X_1, Z_1\}$, POMIS$^+_{2,4} = \{W_2, Z_2\}$, POMIS$^+_{3,4} = \{W_3, Z_3\}$ and POMIS$^+_{4,4} = \{W_4\}$. These candidate sets can then be used as components of an intervention

sequence in the NS-SCM-MAB setting, subject to admissibility and the separately specified support condition.

Scope of the illustration. The IB and local-border calculations above illustrate candidate sets. Calling their unions POMIS$^+$ requires the model-specific support premise of Def. 5.2; the graph alone does not establish it.

G Proofs

Theorem 5.1. *(Existence of Optimal Partial Assignment). Fix a future time t' and an intervention set $\mathbf{X}_{t'} \subseteq \mathbf{V}_{t'} \setminus \{Y_{t'}\}$. Let $\mathbf{D}_{t'} = Pa(\mathbf{V}_{t'}) \setminus \mathbf{V}_{t'}$ be the external-parent boundary. Fix any exogenous conditioning used in the temporal models, and let $\mathcal{H} \subseteq \mathscr{D}(\mathbf{D}_{t'})$ be a nonempty finite set of boundary assignments for which the temporal models are well-defined and the reward expectations are finite. Assume that $\mathscr{D}(\mathbf{X}_{t'})$ is nonempty and finite. For $\mathbf{v} \in \mathcal{H}$, define*

$$g_{\mathbf{X}_{t'}}(\mathbf{v}) = \max_{\mathbf{x} \in \mathscr{D}(\mathbf{X}_{t'})} \mathbb{E}^{\mathcal{M}_{t'}|\mathbf{v}}[Y_{t'} \mid do(\mathbf{X}_{t'} = \mathbf{x})], \qquad G_{\mathbf{X}_{t'}} = \max_{\mathbf{v} \in \mathcal{H}} g_{\mathbf{X}_{t'}}(\mathbf{v}).$$

There exist $\mathbf{W} \subseteq \mathbf{D}_{t'}$ and assignments $\mathbf{w}^*, \mathbf{x}^*$ such that the cylinder $\{\mathbf{v} \in \mathcal{H} : \mathbf{v}[\mathbf{W}] = \mathbf{w}^*\}$ is nonempty and

$$\mathbb{E}^{\mathcal{M}_{t'}|\mathbf{v}}[Y_{t'} \mid do(\mathbf{X}_{t'} = \mathbf{x}^*)] = G_{\mathbf{X}_{t'}}$$

throughout that cylinder. An inclusion-minimal such $\mathbf{W}$ exists.

Proof. The set of pairs $\mathcal{H} \times \mathscr{D}(\mathbf{X}_{t'})$ is nonempty and finite, and all reward expectations in the statement are finite. Therefore the maximum defining $G_{\mathbf{X}_{t'}}$ is attained at some pair $(\mathbf{v}^*, \mathbf{x}^*)$. Take $\mathbf{W} = \mathbf{D}_{t'}$ and $\mathbf{w}^* = \mathbf{v}^*$. The corresponding cylinder in $\mathcal{H}$ consists of the single assignment $\mathbf{v}^*$ and hence is nonempty and attains $G_{\mathbf{X}_{t'}}$ under the fixed assignment $\mathbf{x}^*$. Among the finitely many subsets of $\mathbf{D}_{t'}$ whose cylinders through $\mathbf{v}^*$ are nonempty and attain $G_{\mathbf{X}_{t'}}$ under that same $\mathbf{x}^*$, at least one is inclusion-minimal. This proves the claim. It does not compare $\mathbf{X}_{t'}$ with other intervention sets. □

Why the published arbitrary-model interpretation fails. Let each of two time slices contain a root B_i and the bow $A_i \to Y_i$, $A_i \leftrightarrow Y_i$, with the additional transition $B_1 \to A_2$. Let U_1, U_2 be independent Bernoulli$(1/2)$ variables, independent of the roots B_i, and set

$$A_1 = U_1, \qquad Y_1 = (A_1 + U_1)/2, \qquad A_2 = B_1 \oplus U_2, \qquad Y_2 = (A_2 + U_2)/2.$$

The empty set is a POMIS of the bow graph: in the separate witnessing model $A = U, Y = 1[A = U]$, observation yields reward 1 whereas either constant intervention yields $1/2$. In the displayed additive model, however, for either boundary assignment $B_1 = b$,

$$E[Y_2 \mid B_1 = b] = 1/2 < E[Y_2 \mid do(A_2 = 1), B_1 = b] = 3/4.$$

Thus no boundary assignment makes that graph-level POMIS optimal in this particular model. POMIS asserts existence of a witnessing SCM, not optimality in every conforming SCM. Also, invariance of the optimizer does not imply invariance of the optimal reward value: with $Y = (A + X)/2$, $X = 1$ is optimal for both values of A, but its reward is $(A + 1)/2$.

Proposition 5.1. *(Admissible Time-t Interventions). Every POMIS$^+_{t,t'}$ in Def. 5.2 is contained in $\mathbf{V}_t \setminus \{Y_t\}$.*

Proof. Both $\mathbf{A}_t$ and $\mathbf{B}_t$ in Def. 5.2 are subsets of $\mathbf{V}_t \setminus \{Y_t\}$, so their union is also such a subset. □

Failure of automatic future support (published Proposition G.1). Let each of two identical time slices have roots B_i, C_i and the bow $A_i \to Y_i$, $A_i \leftrightarrow Y_i$. Add transitions $B_1 \to A_2$ and $C_1 \to Y_2$. With $U_2 \sim \text{Bernoulli}(1/2)$ independent of the first slice, define

$$A_2 = B_1 \oplus U_2, \qquad Y_2 = \big(C_1 + 1[A_2 = U_2]\big)/2.$$

For completeness, the first slice can use independent U_1 with $A_1 = U_1$, $Y_1 = 1[A_1 = U_1]$, and independent binary roots B_1, C_1. The empty set is a POMIS of the future bow. The full graph after cutting A_2 has MUCT $\{Y_2\}$ and IB $\{A_2, C_1\}$, so $IB^+_{1,2} = \{C_1\}$ is an allowed raw candidate. If $B_1 = 0$, observation gives $(C_1 + 1)/2$, larger than the value $(C_1 + 1/2)/2$ of either constant intervention

on A_2. If $B_1 = 1$, observation gives $C_1/2$, which is smaller than either intervention. Consequently no assignment to the selected candidate $\{C_1\}$ preserves optimality across all remaining values of B_1. It also cannot force the maximum fixed-set reward, which is 1 and requires $(B_1, C_1) = (0, 1)$. An optimal full-boundary context exists, but the selected IB restriction does not supply it.

Reward nodes must remain non-manipulable. Take identical bow slices $X_i \to Y_i$, $X_i \leftrightarrow Y_i$ and the transition $Y_1 \to X_2$. The full MUCT of Y_2 is $\{X_2, Y_2\}$, whose IB is $\{Y_1\}$. Its time-1 restriction is therefore a reward node, which is inadmissible under Def. 5.1. Cutting Y_1 makes it a root in its own slice, so the old QIB definition gives the empty set; the old composition rule consequently returns the forbidden action $\{Y_1\}$. Filtering admissibility is necessary, and deleting a reward from an IB does not by itself preserve the border's properties.

Proposition 6.1. *(Conditional Composition). Fix the future temporal-model family and intervention set in Def. 5.2. Suppose that an admissible candidate $\mathbf{B} \subseteq \mathbf{C}_{t,t'}$ satisfies that definition's future-support condition, and that $\mathbf{A} \in QIB_t(\mathbf{B})$. Then $\mathbf{A} \cup \mathbf{B}$ is a POMIS$^+_{t,t'}$ relative to those specified choices.*

Proof. By Def. 6.2, $\mathbf{A}$ is an original-slice POMIS. Write $H_t = \mathcal{G}[\mathbf{V}_t]$. In $(H_t)_{\overline{\mathbf{A}}}$, no variable of $\mathbf{B} \setminus \mathbf{A}$ is an ancestor of Y_t. After $do(\mathbf{A} = \mathbf{a})$, the structural equations determining Y_t therefore do not depend on the equations of those variables. Replacing those equations by the constants $\mathbf{b}[\mathbf{B} \setminus \mathbf{A}]$ does not change Y_t. Compatible assignments already agree on $\mathbf{A} \cap \mathbf{B}$. Thus, for every conforming temporal SCM and every fixed boundary context,

$$E[Y_t \mid do(\mathbf{A} = \mathbf{a})] = E[Y_t \mid do(\mathbf{A} = \mathbf{a}, \mathbf{B} = \mathbf{b})].$$

This proves the immediate-reward condition in Def. 5.2. The future-support condition is an explicit premise of the proposition, so that definition applies and yields the conclusion. □

Scope of the correction. The preceding proof establishes a sufficient local composition rule. It does not show that every raw IB restriction has a qualifying future assignment, that every POMIS with specified future support is generated by the graph construction, or that the Cartesian product of locally constructed sets contains all globally optimal intervention sequences. The published unconditional composition result, Proposition G.1, and the dependent exact POMIS$^+$ soundness/completeness guarantee require those missing claims and are not established by the corrected local result. The fixed-plan optimal-value guarantee in Thm. 7.1 instead follows independently from aggregate-reward static reduction.

H Static Reduction Details

The guarantee in Thm. 7.1 needs only forward projection and the static border dominance argument [Lee and Bareinboim, 2018, 2019]. It does not require restoring an arbitrary projected SCM on the original graph.

Forward preservation. Augment the original SCM by $R = \sum_t Y_t$. Substitute the equations of every endogenous variable outside $\mathbf{A} \cup \{R\}$ in topological order, retaining the original exogenous vector and its joint law. Each removed variable is non-manipulable. For any assignment of that exogenous vector and any intervention on $\mathbf{A}$, topological induction shows that every retained variable has the same value before and after substitution. In particular,

$$R^{\mathcal{M}_H}_{do(\mathbf{s})}(\mathbf{u}) = \sum_{t=1}^{T} Y^{\mathcal{M}}_{t,do(\mathbf{s})}(\mathbf{u}).$$

Shared inputs are not replaced by independent copies. The projected graph records all resulting directed and bidirected connections; its permitted inputs can include inputs that a particular model ignores. A source may have more than two retained children.

Lemma H.1 (Static border mixture)**.** *Let $\mathcal{K}$ be a compatible SCM on a finite acyclic mixed graph H whose endogenous variables are $\mathbf{A} \cup \{R\}$. Exogenous sources are independent roots, with all shared uses represented in the graph; each source may have any number of children. Assume every*

domain in $\mathbf{A}$ *is nonempty and finite, and the reward is integrable under every allowed intervention. Fix* $\mathbf{S} \subseteq \mathbf{A}$ *and its assignment* $\mathbf{s}$*, and put*

$$\mathbf{T} = \mathrm{MUCT}(H_{\overline{\mathbf{S}}}, R), \qquad \mathbf{B} = \mathrm{IB}(H_{\overline{\mathbf{S}}}, R), \qquad \mathbf{B}' = \mathbf{B} \setminus \mathbf{S}.$$

Writing $\mu_{\mathbf{s}} = \mathbb{E}^{\mathcal{K}}[R \mid do(\mathbf{s})]$*, one has*

$$\mu_{\mathbf{s}} = \sum_{\mathbf{b}' \in \mathscr{D}(\mathbf{B}')} \mu_{\,\mathbf{s}[\mathbf{S}\cap\mathbf{B}]\,\cup\,\mathbf{b}'} P^{\mathcal{K}}_{do(\mathbf{s})}(\mathbf{B}' = \mathbf{b}') \le \max_{\mathbf{b} \in \mathscr{D}(\mathbf{B})} \mu_{\mathbf{b}}.$$

The assignment inside the mean is to $\mathbf{B}$*, with the common coordinates fixed to the values in* $\mathbf{s}$*. Terms with zero mixing weight are omitted.*

Proof. All ancestors and territory closures in this argument are taken in $H_{\overline{\mathbf{S}}}$. No member of $\mathbf{S}$ belongs to $\mathbf{T}$: a cut vertex has no incoming directed or bidirected edge and cannot be reached by the closure starting at R. The territory is closed under descendants and bidirected connections within the ancestral graph. Its external parents are precisely $\mathbf{B}$. These closure properties give the action/observation exchange separation

$$(R \perp \mathbf{B}' \mid \mathbf{S})_{H_{\overline{\mathbf{S}}\,\underline{\mathbf{B}'}}}.$$

This separation also holds in the expanded DAG when one source has multiple children. Rule 2 therefore gives, for every positive-weight $\mathbf{b}'$,

$$\mathbb{E}[R \mid do(\mathbf{s}), \mathbf{B}' = \mathbf{b}'] = \mathbb{E}[R \mid do(\mathbf{s}, \mathbf{b}')].$$

After fixing the full border $\mathbf{B}$, no directed route from $\mathbf{S} \setminus \mathbf{B}$ to R remains, even when deleting interventions restores their incoming edges. Otherwise take the last $\mathbf{S}$ vertex on such a route. Its suffix lies in $H_{\overline{\mathbf{S}}}$ and must first enter the territory through an external parent in $\mathbf{B}$, which is impossible after cutting incoming edges to $\mathbf{B}$. Rule 3 therefore deletes these remaining interventions, yielding

$$\mathbb{E}[R \mid do(\mathbf{s}, \mathbf{b}')] = \mu_{\,\mathbf{s}[\mathbf{S}\cap\mathbf{B}]\,\cup\,\mathbf{b}'}.$$

The law of total expectation proves the identity, and a probability average cannot exceed its largest summand. This proof uses graph separations and the actual model; it makes no existential choice of a different SCM and imposes no observational positivity assumption. □

Proof of Thm. 7.1. The forward construction produces one projected SCM $\mathcal{M}_H$ with the same aggregate intervention mean for every fixed plan. Apply Lem. H.1 to each such plan. Its dominating border is included in $\mathcal{C}(H, R)$ by the iteration $\mathbf{W} = \mathbf{S}$. Since the algorithm retains every assignment to that border, its maximum is at least the value of the original plan. Maximize over all plans. Conversely, every border is a subset of $\mathbf{A}$ and every assignment retained by the algorithm is feasible, which proves the reverse inequality. The finite nonempty domains ensure both maxima exist. There is at least one iteration, including $\mathbf{W} = \emptyset$, so the returned family is nonempty; all its time components are subsets of the allowed variables in that slice. The same substitution and separation arguments accommodate cross-slice sources and original rewards that are parents of later action variables. No first-order or repeated-topology condition is used.

Proof of Prop. 7.1. For each pair of retained vertices, latent-projection edge conditions are finite reachability questions through removed vertices in the expanded graph. Equivalently, one can eliminate vertices and update directed and bidirected edges. These are polynomial graph operations. An ancestral search followed by closure under descendants and bidirected connectivity also takes polynomial time. The algorithm does this for each of the $2^{|\mathbf{A}|}$ subsets and stores a chronological partition of its border. Its distinct borders are subsets of $\mathbf{A}$, which gives the stated count. Assignment enumeration has the separately stated output size. No tight lower bound is asserted.

The graphical family can overinclude for a prescribed sum. The ordinary static graphical POMIS family is $\{\mathrm{IB}(H_{\overline{\mathbf{W}}}, R) : \mathbf{W} \subseteq \mathbf{A}\}$. Its usual strict-optimality witnesses allow a freely chosen reward equation. The original equation $R = \sum_t Y_t$ is an additional restriction, so such witnesses need not lift to the temporal subclass. For a concrete first-order example with no cross-slice confounding, take two slices whose only within-slice edge is $X_t \leftrightarrow Y_t$, and add $X_1 \to Y_2$. The allowed variables are $\{X_1, X_2\}$. The projected graph has $X_1 \to R$, $X_1 \leftrightarrow R$, and $X_2 \leftrightarrow R$. Its raw static borders are

$\emptyset$ and $\{X_1\}$. Yet in every SCM compatible with the original graph, Y_1 is unaffected by interventions and X_1 is independent of the exogenous inputs in the structural function for Y_2. Thus, with $c = \mathbb{E}[Y_1]$ and $g(x) = \mathbb{E}[Y_2 \mid do(X_1 = x)]$,

$$\mathbb{E}[R] = c + \sum_x g(x)P(X_1 = x) \leq c + \max_x g(x) = \max_x \mathbb{E}[R \mid do(X_1 = x)].$$

Intervening on X_2 has no directed effect on either reward. Consequently observation cannot strictly outperform the other feasible MIS $\{X_1\}$, although $\emptyset$ is a POMIS for the projected graph with a freely chosen scalar reward equation. This distinction does not affect optimal-value coverage, including for prescribed binary action domains.

I Experiment Details

In this section, we provide detailed specifications of the SCMs used in our experiments to ensure reproducibility. For each experiment, the simulation was repeated 200 times using the corresponding SCM. All experiments were run on a dual-socket Intel Xeon Gold 5317 system with 24 physical cores (48 logical threads) at 3.0GHz. These historical runs have not been reproduced in this correction. The released implementation uses independent binary exogenous variables with the probabilities below and Bernoulli surrogate feedback as described in §8.

Task 1. We implement an SCM composed of three time steps $t = 1, 2, 3$ with variable sets $\{X_t, Z_t, Y_t\}$ and structural equations defined to model the propagation of intervention effects over time. The probability distributions over exogenous variables $\mathbf{U}$ are defined as:

$$\begin{array}{llll} P(U_{X_1} = 1) = 0.85, & P(U_{X_1Y_1} = 1) = 0.47, & P(U_{Z_1} = 1) = 0.14, & P(U_{Y_1} = 1) = 0.02, \\ P(U_{X_2} = 1) = 0.80, & P(U_{X_2Y_2} = 1) = 0.55, & P(U_{Z_2} = 1) = 0.14, & P(U_{Y_2} = 1) = 0.01, \\ P(U_{X_3} = 1) = 0.01, & P(U_{X_3Y_3} = 1) = 0.51, & P(U_{Z_3} = 1) = 0.13, & P(U_{Y_3} = 1) = 0.05. \end{array}$$

The structural functions f_V for each endogenous variable V are as follows (where $\oplus$ denotes binary XOR and v is the valuation of its parents):

$$\begin{array}{l} f_{X_1} = U_{X_1} \oplus U_{X_1Y_1} \\ f_{Z_1} = X_1 \oplus U_{Z_1} \\ f_{Y_1} = Z_1 \oplus U_{Y_1} \oplus U_{X_1Y_1} \end{array}$$

(a) $t = 1$

$$\begin{array}{l} f_{X_2} = X_1 \oplus U_{X_2} \oplus U_{X_2Y_2} \\ f_{Z_2} = X_2 \oplus U_{Z_2} \\ f_{Y_2} = Z_2 \oplus U_{Y_2} \oplus U_{X_2Y_2} \end{array}$$

(b) $t = 2$

$$\begin{array}{l} f_{X_3} = X_2 \oplus U_{X_3} \oplus U_{X_3Y_3} \\ f_{Z_3} = X_3 \oplus U_{Z_3} \\ f_{Y_3} = Z_3 \oplus U_{Y_3} \oplus U_{X_3Y_3} \end{array}$$

(c) $t = 3$

Figure 9: SCM definition for Task 1.

Task 2. We conducted the experiment with a two-step SCM defined over variables $\{W_t, X_t, Z_t, Y_t\}$ for $t = 1, 2$, as illustrated in Fig. 4. The structure explicitly models how intervention effects on X_t propagate across time via X_{t+1} and influence downstream outcomes Y_t and Y_{t+1} through intermediary variables Z_t.

The exogenous distribution $P(\mathbf{U})$ is parameterized to highlight the long-term influence of early interventions. The assigned probabilities are as follows:

$$\begin{array}{llll} P(U_{X_1} = 1) = 0.15, & P(U_{Z_1} = 1) = 0.15, & P(U_{Y_1} = 1) = 0.02, & P(U_{X_1Y_1} = 1) = 0.47 \\ P(U_{W_1} = 1) = 0.15, & P(U_{X_2} = 1) = 0.02, & P(U_{Z_2} = 1) = 0.12, & P(U_{X_2Y_2} = 1) = 0.55 \\ P(U_{Y_2} = 1) = 0.01, & P(U_{W_2} = 1) = 0.12. & & \end{array}$$

We define the structural equations in Fig. 10 (where $\oplus$ denotes binary XOR). Notably, the structural function for X_2 includes X_1 as a parent, creating an indirect dependency between early decisions and future rewards through X_2 and Z_2. This design allows us to test the effectiveness of POMIS$^+$ strategies in capturing delayed causal influences that are missed by myopic approaches.

$$f_{W_1} = U_{W_1}$$
$$f_{X_1} = U_{X_1} \oplus W_1 \oplus U_{X_1Y_1}$$
$$f_{Z_1} = X_1 \oplus U_{Z_1}$$
$$f_{Y_1} = Z_1 \oplus U_{Y_1} \oplus U_{X_1Y_1}$$

(a) $t = 1$

$$f_{W_2} = U_{W_2}$$
$$f_{X_2} = U_{X_2} \oplus W_2 \oplus U_{X_2Y_2} \oplus X_1$$
$$f_{Z_2} = X_2 \oplus U_{Z_2}$$
$$f_{Y_2} = Z_2 \oplus U_{Y_2} \oplus U_{X_2Y_2}$$

(b) $t = 2$

Figure 10: SCM definition for Task 2.

Task 3. To clearly illustrate how information from early IB$^+$ and QIB components can propagate across multiple time steps in the absence of the time-slice Markov assumption (Assumption 3.1), we design an SCM inducing Fig. 6. This structure focuses on exposing the long-range effect of early interventions. The SCM contains three time steps $t = 1, 2, 3$ over variables $\{W_t, X_t, Y_t\}$, with unobserved confounding between X_t and Y_t at each step. A bidirected edge between X_2 and X_3 further emphasizes the departure from the time-slice Markov assumption. This configuration enables us to test whether POMIS$^+$ can successfully capture reward-relevant information originating from earlier steps. Conditioning on the middle slice can open the path $X_1 \to X_2 \leftrightarrow X_3$, so this graph need not satisfy the time-slice Markov assumption.

The structural functions f_V for each endogenous variable V are defined as follows:

$$f_{W_1} = U_{W_1}$$
$$f_{X_1} = U_{X_1} \oplus W_1 \oplus U_{X_1Y_1}$$
$$f_{Y_1} = X_1 \oplus U_{Y_1} \oplus U_{X_1Y_1}$$

(a) $t = 1$

$$f_{W_2} = U_{W_2}$$
$$f_{X_2} = U_{X_2} \oplus W_2 \oplus U_{X_2X_3} \oplus U_{X_2Y_2} \oplus X_1$$
$$f_{Y_2} = X_2 \oplus U_{Y_2} \oplus U_{X_2Y_2}$$

(b) $t = 2$

$$f_{W_3} = U_{W_3}$$
$$f_{X_3} = U_{X_3} \oplus W_3 \oplus U_{X_2X_3} \oplus U_{X_3Y_3} \oplus X_2$$
$$f_{Y_3} = X_3 \oplus U_{Y_3} \oplus U_{X_3Y_3}$$

(c) $t = 3$

Figure 11: SCM definition for Task 3.

The probability distributions over exogenous variables $\mathbf{U}$ are defined as:

$$P(U_{X_1} = 1) = 0.82, \quad P(U_{W_1} = 1) = 0.15, \quad P(U_{X_1Y_1} = 1) = 0.52, \quad P(U_{X_2} = 1) = 0.92,$$
$$P(U_{Y_1} = 1) = 0.02, \quad P(U_{X_2Y_2} = 1) = 0.44, \quad P(U_{X_2X_3} = 1) = 0.43, \quad P(U_{Y_2} = 1) = 0.01,$$
$$P(U_{W_2} = 1) = 0.42, \quad P(U_{X_3} = 1) = 0.20, \quad P(U_{X_3Y_3} = 1) = 0.48, \quad P(U_{Y_3} = 1) = 0.05,$$
$$P(U_{W_3} = 1) = 0.41.$$

J Alternative Methods to Approach on NS-SCM-MAB

J.1 Projection-based approach on NS-SCM-MAB

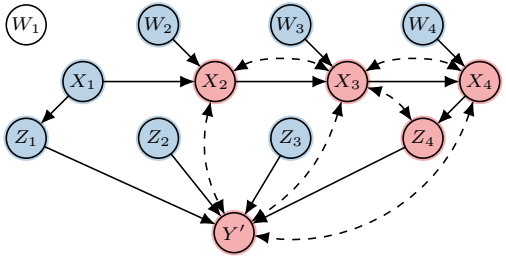


Figure 12: MUCT and IB in the $\mathcal{G}_{[\mathbf{V}\setminus\mathbf{Y}]}$

Lee and Bareinboim [2019] builds on the framework of Lee and Bareinboim [2018] by relaxing the assumption that every variable in the system must be manipulable, while still operating under a stationary setting. Their method uses latent projections (Verma and Pearl [1990], Verma [1992]) to

identify possibly-optimal intervention targets and infer dependency relations among them, even when only partial causal information is available.

A fixed intervention sequence can be represented in the full time-expanded SCM, with a reward node $Y' = \sum_{t=1}^{T} Y_t$. Each non-reward variable retains its time index and interventions are executed at that index. Thus a static representation of a complete episode does not by itself discard temporal order. The construction of Lee and Bareinboim [2019] can be used to remove non-manipulable intermediate variables, while retaining the manipulable variables and Y'.

First of all, we introduce the projection-based approach. In a stationary setting with non-manipulable variables, Lee and Bareinboim [2019] characterize possibly-optimal arms using the latent projection technique [Verma and Pearl, 1990, Verma, 1992]. They begin with a causal diagram $\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$, and define a set of non-manipulative variables $\mathbf{N} \subset \mathbf{V} \setminus \{Y\}$, where Y is the target variable. To construct the latent projection onto the manipulative variables $\mathbf{V} \setminus \mathbf{N}$, they consider an augmented graph $\widehat{\mathcal{G}}$ that explicitly represents all private and shared exogenous roots. They initialize a graph $\mathcal{H} = \langle \mathbf{V} \setminus \mathbf{N}, \emptyset \rangle$, then add edges as follows:

1. A directed edge between V_i and V_j if $V_i \to V_j \in \mathcal{G}$ or there exists a directed path from V_i to V_j where all non-end vertices in the path between there are in $\mathbf{N}$.
2. A bi-directed edge between V_i and V_j if $V_i \leftrightarrow V_j \in \mathcal{G}$; or there exists directed paths from an exogenous root to V_i and V_j in $\hat{\mathcal{G}}$ where all non-end vertices are in $\mathbf{N}$.

Proposition J.1 (Forward preservation under projection)**.** *Let $\mathbf{R} = \mathbf{V} \setminus \mathbf{N}$ include the reward, and let $\mathcal{M}$ conform to $\mathcal{G}$. Substituting the structural equations of the non-manipulable variables yields a projected model $\mathcal{M}_{\mathbf{R}}$ on $\mathbf{R}$ whose retained-variable interventional distributions agree with those of $\mathcal{M}$, for every intervention on $\mathbf{R}$.*

Proof. In topological order, substitute each removed variable's function into its children's functions, retaining the same exogenous sources and their joint law. For every exogenous assignment and every intervention on retained variables, the retained values are unchanged. Integrating over that same law proves the claim. Shared sources must be kept shared. □

Proposition J.2 (Scope of reverse recovery)**.** *Every model in the image of this forward construction has an original model realizing its retained interventional distributions. This assertion concerns the image of the construction, not every SCM compatible with the projected graph.*

Proof. Use the original model defining the image element. Mere compatibility with the projected graph is insufficient to invoke this argument; an arbitrary projected SCM need not belong to that image, particularly with prescribed domains. □

Theorem J.1 (Value preservation for a paired model and its projection)**.** *For the paired models of Prop. J.1 and the same feasible retained interventions $\mathcal{A}$, the expected reward is identical for each $a \in \mathcal{A}$. Hence their optimal values and maximizing fixed assignments coincide whenever the maxima exist.*

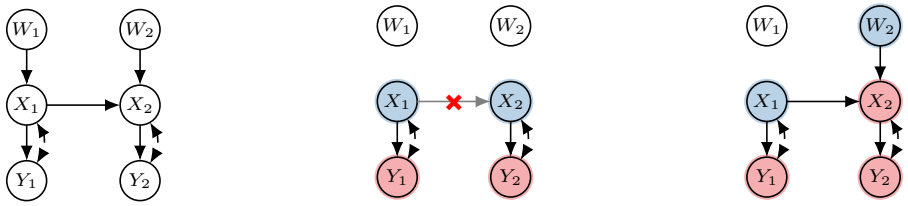
Proof. Apply Prop. J.1 to each intervention and then maximize over the same action family. □

This result does not identify the entire compatible SCM classes or establish an unrestricted graph-level POMIS invariance theorem. The fixed-plan coverage theorem in Thm. 7.1 uses only forward preservation and static border dominance, so it does not require reverse recovery. The published reverse-recovery statement and the argument for POMIS invariance exceeded what the forward construction proves. In particular, a result quantified over freely chosen state spaces cannot automatically be used as a fixed-binary-domain completeness result.

In Fig. 12, the displayed border for the aggregate reward is $\{Z_1, X_1, Z_2, W_2, Z_3, W_3, W_4\}$. This is a graphical candidate; an agreement with a time-indexed construction in this example does not prove general optimality of the latter.

Individual-reward constraints. If the objective is to maximize $\mathbb{E}[Y']$ subject to a bound on $\mathbb{E}[Y_1]$, a representation retaining only Y' cannot directly evaluate that separate reward constraint. Retaining both Y_1 and Y' preserves both required interventional quantities by forward substitution. The time labels of intervention variables can likewise be retained. Therefore temporal interpretability and individual-reward constraints are not intrinsically unavailable to projection. Neither a candidate family selected only for the unconstrained sum nor its temporal decomposition is automatically complete for a constrained problem; feasibility and optimality must be checked against the actual SCM.

J.2 Forced stationary approach on NS-SCM-MAB



(a) $\mathcal{G}$ (b) $\mathcal{G}_{\overline{X_1,X_2}}$ (c) $\mathcal{G}_{\overline{X_1,W_2}}$

Figure 13: (a) Non-stationary causal diagram $\mathcal{G}$ and (b, c) mutilated graphs

Intervening on transition variables can simplify the dependence of rewards on past actions, but deleting a single edge does not in general make the reward stationary.

Definition J.1 (Block)**.** An intervention $do(\mathbf{F}_t = \mathbf{f})$ blocks a transition edge if the edge enters a variable of $\mathbf{F}_t$ and is therefore removed in $\mathcal{G}_{\overline{\mathbf{F}_t}}$. Other routes from earlier slices to Y_t may remain.

Definition J.2 (Forced stationarity)**.** Under corresponding sets $\mathbf{F}_t$ across time and a fixed aligned value $\mathbf{f}$, forced stationarity means that $P(Y_t \mid do(\mathbf{F}_t = \mathbf{f}), I_{<t})$ is the same aligned distribution for all t and all admissible past intervention histories $I_{<t}$. The histories refer to interventions, not conditioning on realized observations.

Lemma J.1 (Sufficient conditions for forced stationarity)**.** *Suppose that, after $do(\mathbf{F}_t = \mathbf{f})$, no variable in an earlier slice is a directed ancestor of Y_t. Suppose also that the remaining ancestral structural equations and the marginal law of their exogenous inputs are identical under time relabeling, using the same $\mathbf{f}$. Then the reward distributions in Def. J.2 agree.*

Proof. Evaluate Y_t in its post-intervention ancestral submodel. It has no input from earlier endogenous variables, and hence no dependence on earlier interventions. The aligned structural equations, fixed values and exogenous marginal law are the same in each slice, so their induced reward laws agree. □

These hypotheses include both removal of every historical causal route into the reward and equality of the remaining mechanisms and noise laws. Blocking $Z_{t-1} \to Z_t$ alone, or having identical graph topology alone, supplies neither condition. For example, $Y_1 = Z_1 \oplus U_1$ and $Y_2 = 1 - (Z_2 \oplus U_2)$ with identically distributed Bernoulli(p) noise, $p \neq 1/2$, have the same within-slice graph but unequal reward laws under the same fixed Z; deleting a transition into Z_2 does not repair this difference.

In Fig. 13, fixing X_2 removes the shown $X_1 \to X_2$ transition. Stationarity requires the additional conditions above. Fixing W_2 leaves that transition intact. Restricting attention to interventions that block transitions may exclude useful alternatives, but their relative reward values require the SCM parameters. The graph alone does not certify that one of these interventions is better.

K More Discussions

K.1 Relaxing Natural Characteristics on Temporal Structure

Repeated within-slice topology and first-order transitions are modeling restrictions. Definitions of raw border slices can also be applied to a general known temporal graph, including graphs with cross-slice confounding, by using the full unrolled graph. This observation does not establish an exact POMIS$^+$ characterization or a global optimality guarantee. If there are no transitions, no future support is

needed; the absence of transitions is a stationary special case when the aligned mechanisms and noise laws are also identical. It is not a reason to omit any time step from an intervention sequence.

K.2 Non-Stationarity in Causal Bandits vs. Causality in Non-Stationary Bandits

It is useful to distinguish our formulation from alternative approaches that combine causality and non-stationarity in opposite directions. Our work builds upon structural causal bandits (SCM-MABs) [Lee and Bareinboim, 2018, 2019], which assume a fixed underlying causal graph and utilize it to guide interventions. The temporal SCM formulation models how actions can alter later reward laws. Related non-stationary causal-bandit formulations, including those of Nourani-Koliji et al. [2023], Huang and Wu [2024], address different assumptions and statistical tasks. The distinction concerns the formulation and available structural information; it is not a claim that non-stationary bandits had never been studied with causality. Knowing the temporal graph alone does not validate the original future-border restriction. The fixed-plan coverage in Thm. 7.1 instead uses the aggregate static construction and its separate proof.

K.3 Composition of temporal rewards

An intervention sequence and the causal responses of each time-indexed variable determine the expected cumulative reward in a sequential decision problem. Under a fixed intervention plan, external parents and shared latent boundary variables are generally random. A conditional temporal reward at one realized boundary cannot be substituted for its unconditional reward without averaging. For each t, let B_t collect the external-parent and shared-latent boundary values, and for this fixed plan define $K_t^a(\mathbf{a}_t, b)$ as the conditional mean of Y_t in the intervened SCM given $B_t = b$. Its local-noise law is the conditional law induced by that same intervened SCM, including any cross-slice confounding. The boundary law is that induced by the preceding interventions. The following is an expectation identity, not an optimization theorem.

Proposition K.1 (Expected reward composition)**.** *For any fixed intervention plan $a = (\mathbf{a}_1, \dots, \mathbf{a}_T)$ with finite reward expectations,*

$$\mathbb{E}^{\mathcal{M}}\left[\sum_{t=1}^{T} Y_t \mid do(a)\right] = \sum_{t=1}^{T} \int K_t^a(\mathbf{a}_t, b)\, dP^{\mathcal{M}}_{do(a_{<t})}(B_t = b).$$

For discrete boundaries, replace each integral by the corresponding probability-weighted sum. The outer expectation may be omitted only when the boundary is fixed by the plan, or the conditional mean is constant over its induced support.

Proof. Linearity of expectation separates the reward sum. Temporal ordering makes the boundary law independent of the current and later interventions, and later interventions cannot affect Y_t. Apply the law of total expectation under the intervened SCM at each time t. □

For instance, an unmanipulated Bernoulli$(1/2)$ boundary B_2 and $Y_2 = B_2$ give an unconditional mean $1/2$, while the two realized-boundary means are 0 and 1. The averaging step cannot generally be dropped. In the absence of cross-slice confounding, the conditional kernel can be evaluated in the fixed-boundary temporal submodel with its appropriate local noise law.

Corollary K.1 (Local optimality does not imply global optimality)**.** *Optimizing immediate reward in each realized temporal model need not optimize the expected sum of rewards over the full horizon.*

Proof. Let both slices contain binary X_t and reward Y_t, with $Y_1 = \frac{1}{4}X_1$ and $Y_2 = \frac{3}{4}(1 - X_1) + \frac{1}{4}X_2$. Both within-slice graphs are $X_t \to Y_t$, and the sole transition is $X_1 \to Y_2$. Greedy choices $X_1 = X_2 = 1$ yield total reward $1/2$, whereas the plan $X_1 = 0, X_2 = 1$ yields 1. Both respect temporal ordering and rewards lie in $[0, 1]$. □

This corollary illustrates that local optimality under temporal models does not guarantee global optimality in the full SCM. Due to the interdependencies between time steps, early interventions may have long-term consequences that are not captured by myopic optimization. Therefore, joint planning across time is necessary to achieve globally optimal decisions. The conditional composition rule in Prop. 6.1 does not by itself solve that planning problem.

Changes

2026-10-25. Unofficial AI-corrected version of Yeahoon Kwon, Yesong Choe, Soungmin Park, Neil Dhir and Sanghack Lee, *Non-Stationary Structural Causal Bandits*, NeurIPS 2025. This final page records changes made in this PDF. Original publication: `https://openreview.net/forum?id=F4LhOqhxkk`.

These proposed corrections were prepared with AI assistance. **Author review pending.** This file is not a publisher-issued correction and does not indicate coauthor approval.

Location	Before → After	Assumptions and limits
§5, Thm. 5.1; App. G	Theorem 5.1 and its proof are replaced: instead of arbitrary-model POMIS optimality, the result gives best-context attainment for one fixed intervention set. Remark 5.1 is removed.	Finite, nonempty context family and action domains; finite expectations. No comparison with other target sets or guaranteed controllable support.
§6, Prop. 6.1; App. G	Raw IB slices were asserted to supply future support automatically; QIB used an incorrect universal quantifier. Composition now requires separately verified future support, original-slice POMIS membership and immediate-reward preservation.	Support depends on the specified future model family and target set. Raw borders can contain forbidden reward nodes. Published Prop. G.1 is withdrawn.
§7; App. H	The future-border restriction can miss a cumulative optimum ($11/8$ versus $13/8$). Algorithm 1, Theorem 7.1 and its proof are replaced by aggregate static reduction and fixed-plan value coverage. Algorithm 3 is removed; Proposition 7.1 and Lemma H.1 are added.	Existing static results apply with non-manipulable rewards, finite full action domains and all subsets/assignments allowed. Exact POMIS$^+$ enumeration, exact minimality and adaptive-policy optimality are not recovered.
§3; App. J.2	Stationarity now compares the same aligned arm. Equal graph topology or deletion of one transition did not ensure stationarity; sufficient mechanism and history-separation conditions are added.	The first-order setting explicitly excludes cross-slice confounding; paths can still span many slices. The two-slice planning sufficiency claim is removed.
App. J.1	Time labels can be retained under projection. Theorem J.1 and its proof are replaced by value preservation for paired original/forward-projected models; reverse recovery is restricted to this image.	No equality of entire SCM classes or unrestricted graph-level POMIS invariance is claimed. Individual-reward constraints require retaining their outcomes.
App. K.3, Prop. K.1	Realized-boundary means were substituted for unconditional rewards. The composition identity now averages over the boundary law induced by prior interventions.	The kernel uses the conditional noise law of the same intervened SCM. This expectation identity does not prove optimality.
§8–9; App. I	Feedback is described as the released Bernoulli$(\mu(a)/3)$ surrogate over fixed sequences. Candidate state can leak between tasks; retained learning curves await reproduction. Task 3's myopic family also contains an optimum.	Reported values and curves are retained. The public runner bug can alter exploration; the exact published invocation and result arrays are unverified.
Abstract, §1–4, 10; App. I	Claims depending on the replaced guarantees are adjusted. Confounding/feedback, indices, reward-set notation, weak dominance, an illustrative sequence and task equations are corrected. Original Eq. (3) is removed; regret Eq. (4) becomes (3).	The three replaced theorems keep their original numbers. Task 2 uses slices $1, 2$; Task 3 uses X_1 in f_{Y_1} and $P(U_{Y_1} = 1) = 0.02$.

Experimental scope. The 200-repetition learning experiments were not rerun; all reported numerical values and figure data are unchanged. Exact checks over all 729 fixed assignments per released SCM found the full-space optima 2.117260, 1.386304 and 1.903800 in the replacement static families (59, 28 and 64 arms). These checks do not reproduce learning curves. The retained curves concern the original temporal method, not the replacement algorithm.

Cite the original paper. If using this version, add the note *Unofficial AI-corrected version dated 2026-10-25 consulted*; include its URL only if a public version is available.

AI review sign-off: gpt-6.1-sol · 2026-10-25.